Imperial College London

EEERover – Engineering Group Design Project

ELEC40006 – Electronics Design Project 1 (Summer 2022-2023)

Team Fyrryx

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by

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Abstract

The rover was built to navigate a specific environment and detect Alien life. It had to detect the infrared signal emitted from the alien which corresponds to its age and the radio waves which relate to its name. In addition, the rover had to determine the direction of the Alien's magnetic field – positive, negative or neutral. This had to be achieved via a remotely controlled interface (XBOX controller) in a short amount of time by manoeuvring quickly and efficiently around the arena. The front-end program had to continuously output data onto a graphical user interface by communicating through the UDP protocol. Consideration was taken to the chassis design to ensure it contained all the sensor circuitry and was lightweight so that it did not exceed the predetermined weight limit. As a result, the design of the rover could be split into two main branches: controlling and sensing. To allow the sensors to communicate with the board, interfacing took place between the Adafruit and Orangepip boards which increased pin availability. Testing took place through LTSpice circuit simulations and independent experiments were conducted to determine whether each submodule met the desired specification. Conclusions were drawn that the user interface was accessible, and the remote control worked reliably with low latency. Additionally, each of the sensors worked to a high degree of sensitivity with a suitable range to detect the input signals.

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1. Introduction

The motivation of this group design electronics project is to "design a remotely controlled rover that can explore a remote planet" [1] [1]. The rover must be able to identify and detect alien life within the designated environment and determine each alien's, name, age, and direction of the magnetic field. This can be achieved using analogue and digital methods along with RLC circuits and Arduino programming. The system requirements that have been chosen to be implemented within the rover include being able to navigate the arena quickly and efficiently, detect each alien and its respective parameters and output the data collected onto a graphical user interface that can be easily interpreted.

Subsequently, five main submodules were identified to break the problem down into more manageable sections that can be worked on and combined to create a finished product. These include name, age and magnetic field detection as well as remote controlling and design of the chassis. The design cycle chosen is an iterative cyclic process which involves planning, developing, creating and evaluating at each stage during the technical development of the rover (Appendix 1 - [2] [2]). The budget of £60 must not be exceeded and can be used to buy specific components from online suppliers – cost analysis (Appendix 2).

<u>Arena Analysis</u>

The arena has been designed to have a smooth floor with aliens dotted throughout the environment. In addition, there are certain objects including RGB lights encased inside a bowl that must be manoeuvred around and rocks that transmit various signal frequencies (figure 1.1). Therefore, these rocks aim to disrupt the sensing of the magnetic field direction, infrared signal and radio signal. A method implemented within all sensors to overcome this obstacle is to ensure that the range of the sensors is small enough to ensure that the rover is required to get close to the alien it is sensing avoiding interference from the rock objects. The smooth floor of the arena ensures that the rover can easily navigate the arena without getting stuck. In addition, part of the arena contains an alien which is sensitive to a weight limit. If exceeded the light on the scale will turn red and prevent the surrounding aliens from transmitting their signals (figure 1.2).

Figure 1.1: Arena Environment

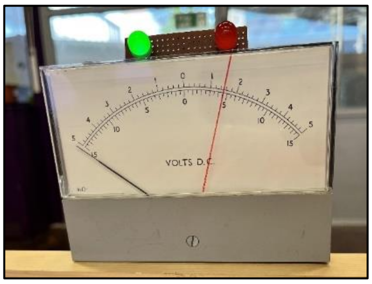

Figure 1.2: Weight Scale

Figure 1.3: Alien

2. Name Detection

2.1 Theory

2.1.1 Aims

This sensor should determine the name of each alien in the arena. The name is communicated via radio, encoded in UART packets using ASCII codes. It is modulated with “two-level amplitude-shift-keying” and has a carrier frequency of 61kHz [1]. The brief also specifies the data rate as 600 bits per second and each name as consisting of a hashtag symbol followed by three more characters.

2.1.2 Tasks involved

This problem was broken down into five main steps:

1) Convert radio waves into a signal
2) Amplify the signal amplitude
3) Demodulate the signal
4) Convert into a binary signal
5) Decode into hexadecimal

2.2 Design Specification

Table 2.1 identifies the attributes specific to the ‘name’ sensor that are necessary to meet the brief’s requirements.

Feature	Quantitative specification	Qualitative specification
Detect Radio Signal		Returns a hexadecimal integer corresponding to the name of the Alien
Compact	<16cm width < 160mm length	Must fit within the width of the chassis and allow room for other sensors.
Range	11cm distance	Large enough to not be inconvenient for the operator, but small enough not to be affected by other signals such as the rocks in the arena.
Aesthetics	0 exposed wires	Exposed wires can cause difficulty during debugging. A well-designed circuit is easier to debug, and less likely to have issues with loose connections.
Reliability	Operates correctly for 15-35°C >1 component purchased	The lab can get hot in summer, so the sensor must be stable within this extended range. Backup components should be ordered from reliable manufactures in case of breakage/defective parts.
Time Constraint	<4 weeks	Ensure purchased components have a short delivery time/are not on backorder. Order components early in the project timeline
Budget Constraint	<£10	Budget must be shared across all submodules

Table 2.1 Name Detection Design Specification

2.3 Experimental Method

2.3.1 Conceptual Designs

Task 1 – Convert radio waves into a signal

To complete this task, a resonant circuit can be built, comprising of a capacitor and a tuned coil antenna [1]. The circuit should be tuned to the carrier frequency of 61kHz to get the maximum amplitude for the resulting signal.

Figure 2.1 shows the tuned circuit. A restriction on the inductor was that it had to fit between the wheels of the rover, as demonstrated in Table 2.1, so it was built with a diameter of 12cm. The inductor was built to 55µH using the RC meter, so a capacitance of 124nF was required for tuning at 61kHz. As the inductor coil was custom-made, it was possible to fine-tune the inductance to make it easier to find suitable capacitors from the finite values available. Thus, a capacitance of 122nF was used and the inductance was increased to 57µH. The circuit was subsequently tested using a Picoscope: the resulting waveform can be seen in Figure 2.2.

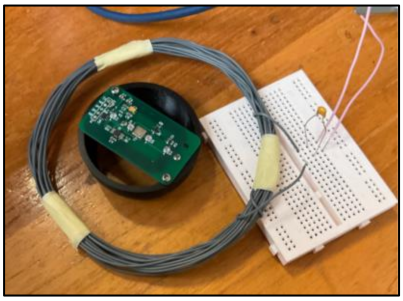

Figure 2.1: Resonant Circuit

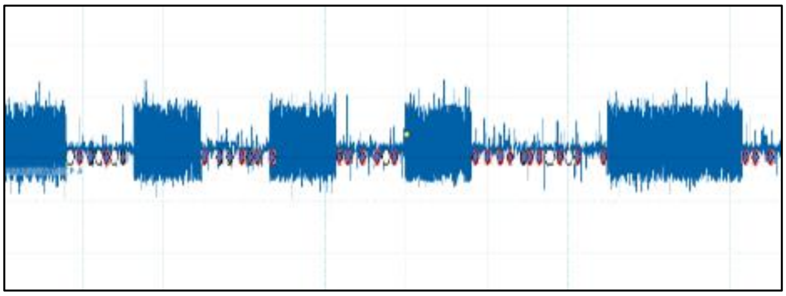

Figure 2.2: Picoscope Signal

Task 2 – Amplify the signal amplitude

As the amplitude of the output signal was small at 150mV, amplification was required before demodulation [1] [1]. This was because the subsequent envelope detection included a diode, which dropped 0.7V across it. Thus, the signal amplitude needed to be larger than this to retain a signal after rectification. A gain of 40dB was chosen to exceed this requirement.

It was decided to amplify using an OpAmp rather than a single-transistor amplifier for circuit simplification and better performance, including higher gain and temperature stability [3] [3], as required in Table 2.1. There was little difference in inverting/non-inverting configurations in testing, provided appropriate voltages were applied to the rails, and it was not necessary to preserve a non-inverting signal, so either could be chosen for this purpose.

Analog Filter Wizard [4] [4] was used to design potential circuits before LTSpice [5] [5] was used to simulate and test the schematics. Larger capacitor values were favoured to avoid issues with parasitic capacitances in breadboards which were used for initial circuit testing.

As significant noise was observed (see Figure 2.2), it was concluded that an active filter should be used in the amplification stage to prevent undesired noise amplification and to accomplish both amplification and filtering in one circuit. The options were low-pass or band-pass, as the noise was high frequency.

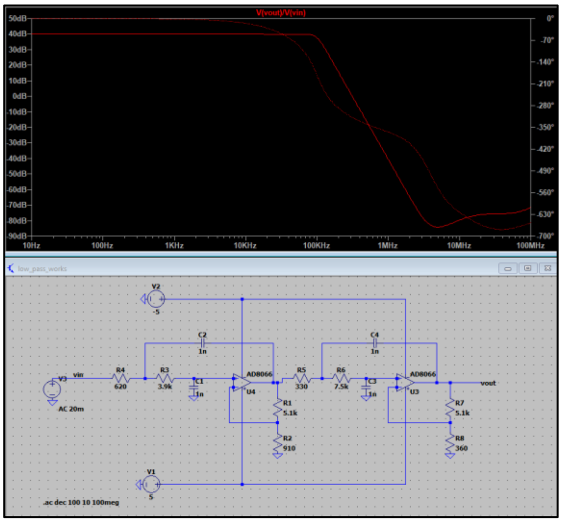

Figure 2.3: Active Low-Pass Circuit

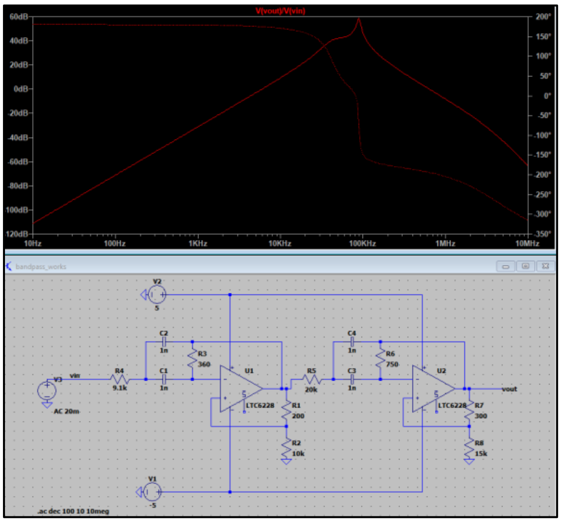

Figure 2.4: Active Band-Pass Circuit

Eventually, the low-pass design was chosen due to its lower sensitivity to resonance. These circuits can be compared in Figures 2.3 and 2.4. A corner frequency higher than 61kHz was chosen to allow for discrepancies in the physical circuit.

The minimum gain-bandwidth product of the OpAmp for "good" performance, and attenuation at high enough frequencies was 16MHz [4] [4]. Using the EEE-recommended suppliers to satisfy Table 2.1, the most cost-effective, suitable OpAmp with next-day delivery (Table 2.1) was ordered from OneCall, the AD8056 [6] [6].

While testing with the tuned circuit, it was observed that the circuit oscillated when power was supplied to the amplifier. This was unforeseen in the LTSpice simulation as the simulation used a voltage source rather than a resonant circuit for the input signal, which was not a good model in this case. As the oscillation was a result of feedback to the resonant circuit, this was fixed by removing the feedback capacitor from the first stage. As a result, the gain decreased, but this was increased again by changing resistor values. The updated circuit can be seen in Figure 2.5. The roll-off order was less than before, but this was not a problem as the noise was still reduced to a satisfactory level. The consequential gain of the amplifier was 32.7dB, resulting in an output signal peak-to-peak voltage of 6.8V (Figure 2.6), which was large enough for the subsequent envelope detector.

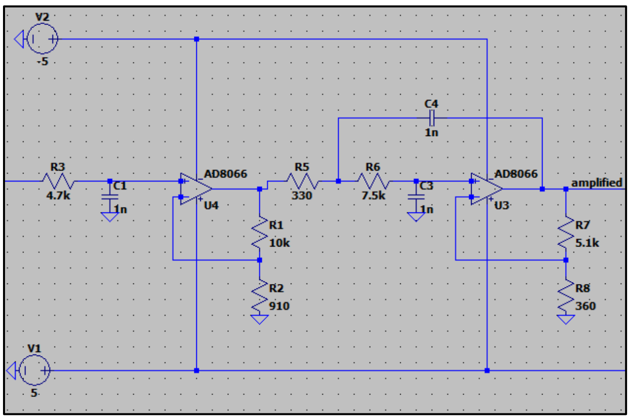

Figure 2.5: Amplifying Stage

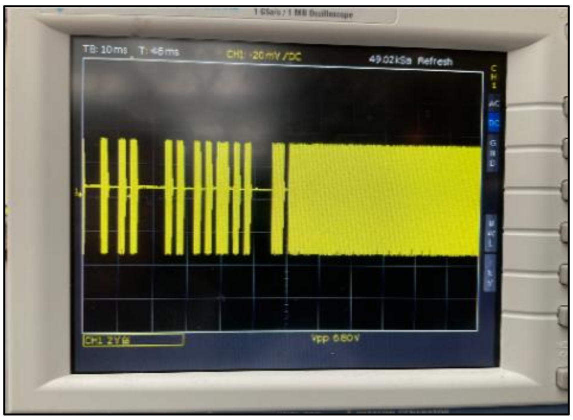

Figure 2.6: Amplified Signal

Task 3 – Demodulate the signal

It was necessary to use envelope detection to rectify the signal because the microcontroller would not be able to sample the signal at its carrier frequency [1] [1]. This circuit detects the 'envelope' (peak amplitude) of the modulated wave in order to obtain the initial signal [7] [7]**.** Due to the previous amplification, a precision rectifier was not required as the amplitude was greater than 0.7V. A simple envelope detector consists of a diode, capacitor and resistor as shown in Figure 2.5 [8] [8]. As 'B', the modulating signal, was 6.8V after amplification, a 100nF capacitor and 1kΩ resistor were used for the circuit.

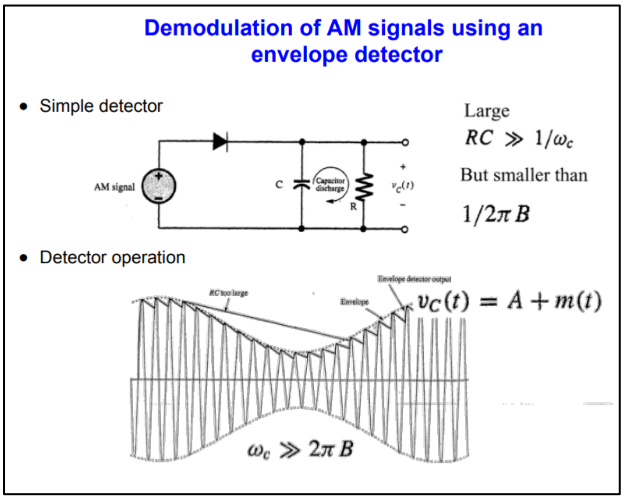

Figure 2.6: Envelope Detection

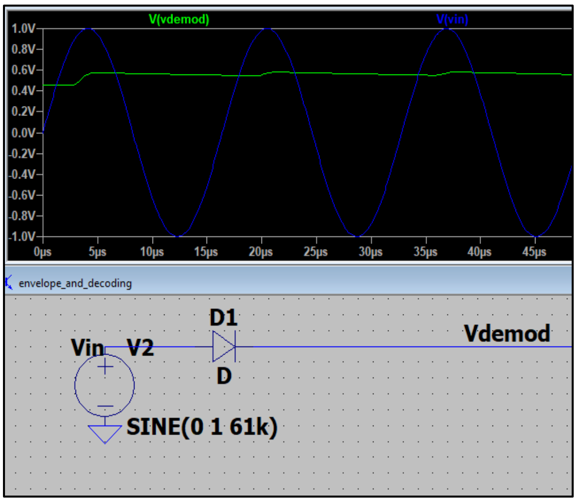

Figure 2.7: Diode-Only Rectification

Rectification using only a diode was trialled using LTSpice (Figure 2.7), but the demodulated signal was biased to 0.5V and there was little difference in amplitude between the 'on' and 'off' signal amplitudes, so it was decided that decoding should include the low-pass filter. Figure 2.8 shows the demodulated signal as seen on the oscilloscope. This circuit met the design specification in Table 2.1 well, as it was compact, no external parts were required, and debugging was simple with such few components. There was again significant noise, so this was considered when designing the demodulation part of the circuit.

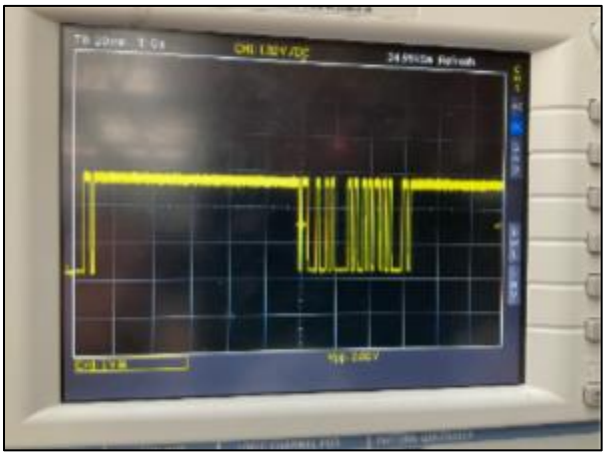

Figure 2.8: Demodulated Signal

Task 4 – Convert into a binary signal

Amplitude-shift-keying represents logic 0 with an amplitude of zero and logic 1 with an amplitude larger than zero [1] [1]. A comparator compares a signal amplitude to a threshold voltage and therefore was suitable for converting the signal into a binary one. It was considered to purchase a Schmitt-Trigger as they implement hysteresis using positive feedback. A comparator using hysteresis allows the reference to depend on the output state, by including both an upper and lower threshold voltage [9]. [9] This can lead to 'cleaner' switching transitions. Schmitt-Triggers were also very cheap, more compact and less susceptible to loose connections than a custom circuit, satisfying the specification in Table 2.1. The 74HCT14 Schmidt-Trigger was ordered [10] [10]. An oversight was that this component did not have an input for a reference voltage, and therefore it was not possible to control the threshold value. Instead, a custom Schmitt-Trigger was built using a lab-provided LT1366 OpAmp. The initial schematic is shown with the preceding envelope detector in Figure 2.9, and the oscilloscope signal in Figure 2.10.

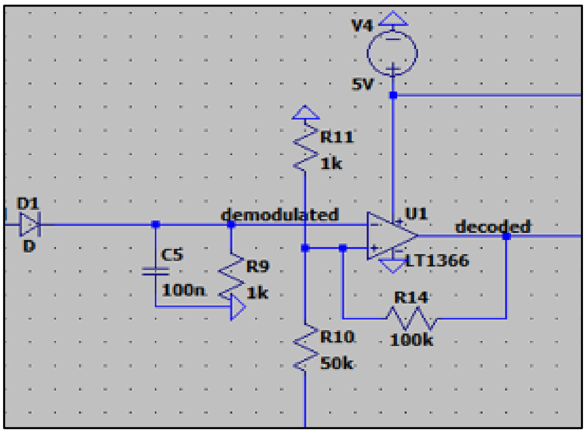

Figure 2.9: Envelope detection and binary conversion schematic

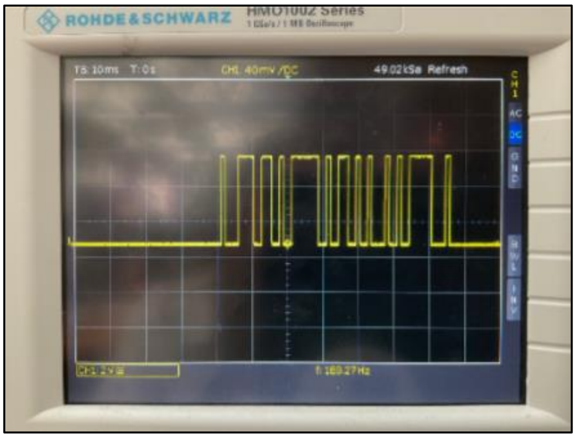

Figure 2.10: Envelope detection and binary conversion waveform

It was observed that the signal was inverted by the Schmitt-Trigger, so a NOT-gate was added using an NPN-transistor and two 50kΩ resistors to facilitate UART decoding. It was observed that with the threshold voltage at such a small value, the signal was HIGH for longer than it should have been (Figure 2.11). This would have been an issue during decoding as the measured baud rate of the signal was over 800 bits/second, whereas it should have been 600. For successful transmission of serial data, the sent and received signals must have a baud rate within 10% of each other [11]. This was easily adjusted using new values of 1k and 1.5k: Figure 2.12 shows the successful result in comparison with the amplified signal.

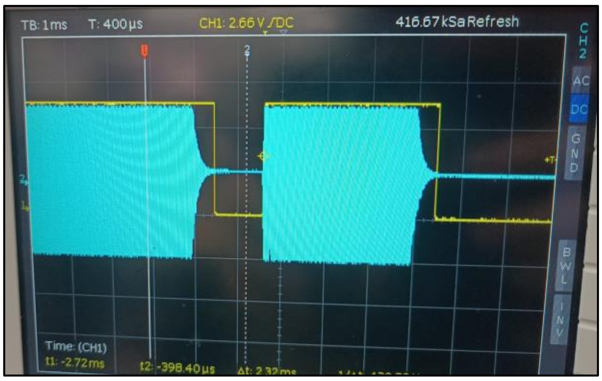

Figure 2.11: Incorrect Threshold Voltage

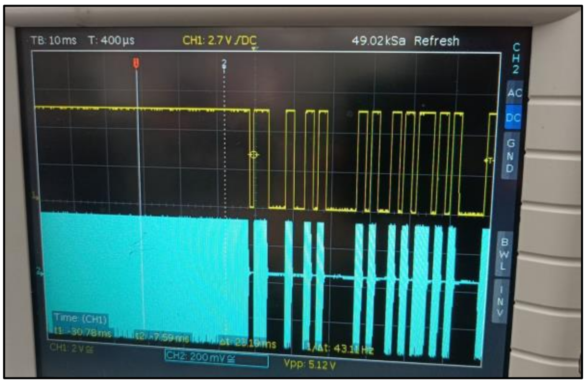

Figure 2.12: Correct Binary Signal

Task 5 – Decode into hexadecimal

The decision to use the Orangepip for sensing (see Section **LOLEZIO INTEGRATION**) was beneficial as the code did not have to be non-blocking, as the motors and other sensors would not be affected. This simplified the required code, as it was possible to use the method 'readStringUntil'. The code used for isolated testing of name is displayed in Figure 2.13.

namecode.ino.ino

```


void setup() {
  //Initialise pin modes and serial
  pinMode(0, INPUT);
  Serial.begin(600);
}

// uses UART to automatically decode the name
String get_name() {
  String name = Serial.readStringUntil('#');
  return name;
}

//If data is available, calls get_name and returns result to the serial monitor
void loop(){
  if(Serial.available()){
    String alien = get_name();
    if(alien.length() == 3){  //reduces noise
      Serial.println(alien);
    }
  }
}

```

Figure 2.13: Name Detection Code

Using the above code and the complete circuit below (Figure 2.14), it was possible to decode Alien 36's name as 'Kay' (more detail in Section 2.4: Testing and Results).

2.3.2 Final Designs

Figure 2.14 shows the complete 'name detection' schematic, and the physical circuit is displayed in Figure 2.15.

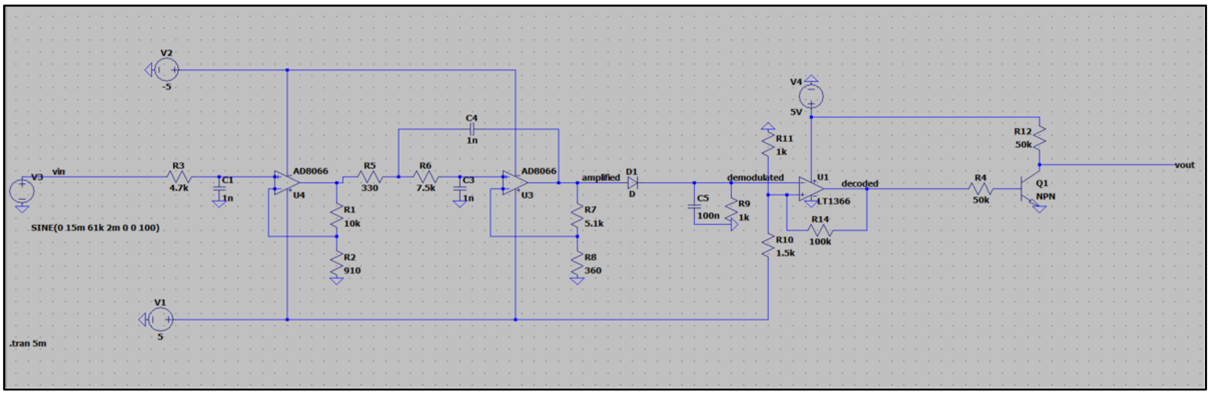

Figure 2.14: Name Detection Schematic

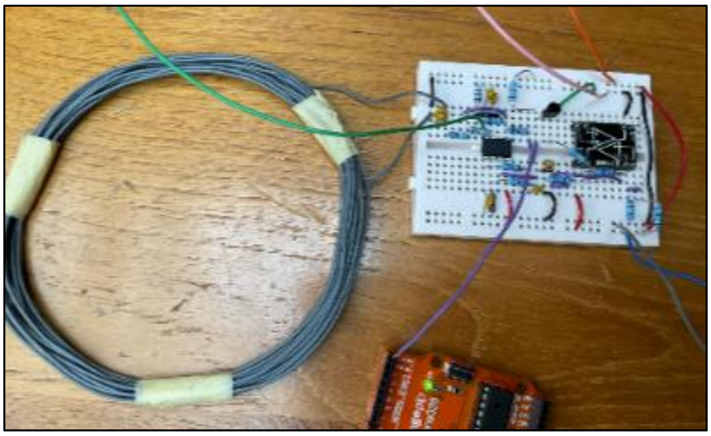

Figure 2.15: Name Detection Circuit

The Veroboard circuit can be seen in Section **LOLEZIO INTEGRATION**

2.4 Testing and Results

Alien Name Detection

The circuit was tested on multiple test aliens from the lab, as shown in Table 2.2.

Alien Number	Name
36	Kay
30	Cai
38	Geo
23	Eli

Figure 2.2: Name Detection Results

Figure 2.16 shows the successful detection of names with an alien in 'CYCLE-NAME' mode.

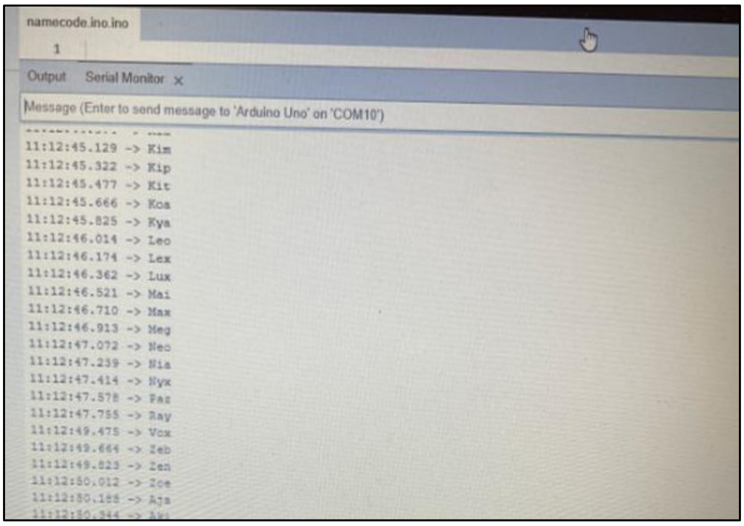

Figure 2.16: Name Detection Circuit

Range Testing

The range at which a name could still be detected was 18cm from the inductor to the alien's base. This is sufficient as the height of the inductor-holder on the chassis is 11cm above ground. This satisfies the range requirement in Table 2.1.

2.5 Conclusion and Evaluation

One challenge that was faced during the process of designing this sensor was that some incorrect information was provided by the client about the carrier frequency. This was simple to resolve, as it was possible to re-tune the resonant circuit and redesign the amplifying stage. Another issue was that the delivery time of the OpAmp was slightly longer than stated on the website. While waiting for it to arrive, testing of the amplifier circuit was attempted using the lab provided LT1366. Of course, it was quickly realised that the gain bandwidth product of this OpAmp was not nearly high enough. This was fine as the OpAmp had been ordered with plenty of time to spare, in accordance with the specification. In the meantime, progress was made on the subsequent tasks for this sensor.

The ‘Name Detection’ sensor satisfied the requirements outlined in Table 1.2. It was compact due to the use of the Veroboard, and the size of the inductor was sufficient for the rover size. The Veroboard also led to a neat, clean circuit which reduced the risk of loose connections and exposed wiring. The range was larger than required, which allowed more freedom for positioning the rover near an alien. All purchased components could operate in the required temperature range. Only £6.07 was spent out of the allocated £10 budget. As a result, this sensor is successful in meeting the brief requirements.

3. Age Detection

3.1 Theory

This section aims to determine the age of each alien. The "alien emits infrared radiation at a wavelength of 950nm" and pulses with a frequency range of 135-1000Hz which must be detected using a designed sensor circuit [1]. The period of the alien calculated correlates to the age of the alien in decades.

3.1.1 Detecting the Infrared Radiation

Initially, a photodiode provided by the department was tested and found that it was sensitive to both infrared and visible light at the same time. A photodiode works by converting the incident light into electrical energy when operating in reverse bias conditions [13]. The output current of a photodiode depends on the light intensity received but it cannot distinguish between whether the light is visible or infrared which has a larger wavelength. As a result, other alternatives to a basic photodiode were explored (Table 3.1).

Method	*Advantage*	*Disadvantage*
Infrared Filter Paper	• Allows infrared light to pass and blocks visible light. • Cheap alternative because this would modify the present photodiode into an infrared diode	• Difficult to purchase from online suppliers in small quantities. • Reliability is unknown
Infrared Receiver (38KHz)	• Can detect the infrared signal at a specific frequency	• The receiver contains a detector that looks for a modulated infrared signal at a specific frequency and demodulates it. This is not suitable as the frequency of the pulse is unknown [12].
Infrared Diode (850nm)	• Detects the infrared light. • Contains a daylight filter to block visible light. • Cheap and has a high sensitivity	• The peak sensitivity wavelength of the sensor is 880nm which is less than the required 950nm. Hence, it may be less sensitive to the radiation emitted by the aliens but not significant to notice.

Table 3.1: Alternative IR Sensor Choices

After evaluating different methods, buying an infrared diode from the online supplier, OneCall QSD123 Phototransistor [14], was the best option due to its reliability and ensuring the correct radiation was received. This is important because, within the arena environment, there will be additional objects that will emit light of different frequencies causing interference.

3.1.2 Converting Current into a Voltage

The output of a photodiode is in the form of a current. This needs to be converted into a voltage. The first method is by connecting the photodiode in reverse bias and passing the reverse bias current through a resistor to obtain a voltage (figure 3.1). Alternatively, figure 3.2 shows a transimpedance amplifier

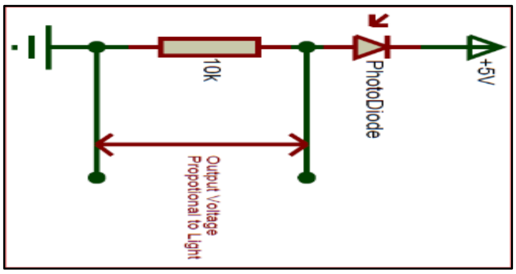

Figure 3.1: Photodiode and Resistor Circuit [29]

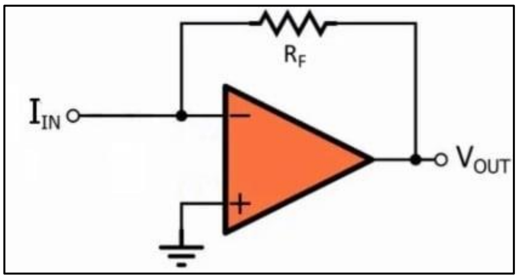

Figure 3.2: Transimpedance Amplifier [15]

which converts a current into a voltage and simultaneously amplifies the voltage signal [15]. The disadvantage of this method is that noise could also be amplified which would disrupt the desired signal. Therefore, the first method was chosen.

3.1.3 Tasks Involved

Five main tasks have been identified when trying to determine the age of each alien.

1) Detecting the infrared signal using an infrared photodiode and converting the current into a voltage.
2) Filtering the signal using a low pass filter to obtain the frequency in the range of 135-1000Hz.
3) Amplifying the signal to give a larger output waveform.
4) Converting the analogue signal to a digital signal.
5) Programming using the Arduino IDE to calculate the period of the pulse and hence the age.

3.2 Design Specification

Feature	Quantitative specification	Qualitative specification
Detect Infrared Signal	Provides a numerical value for the period of each Alien (Alien's Age)	This should detect the signal over a short period; accurately and efficiently. Determine the period of the signal which has a frequency between 135-1000Hz.
Compact	160mm x 160mm	The area the circuit covers should be small enough to ensure that the overall design is compact and fits on the rover chassis.
Range	5cm Distance	The signal strength decreases as the range increases. The range cannot be too large to avoid detecting the wrong object.
Aesthetics	0 exposed wires	The rover should also be well-designed to ensure that exposed wires are not excessive in length. This will make it easier to debug if there are any problems and provides a compact design.
Reliability	Operating temperature range: 15-35°C	The age detection sensor circuit must be able to work over a larger operating temperature range without the output signal being distorted because the labs can get hot.
Quantity	Multiple components (x3)	Multiple sensors should be bought in case one breaks.
Time Constraint	<4 weeks total	Order components with a short delivery time (e.g., components in stock for next-day delivery).

Table 3.2: Age Design Specification

3.3 Experimental Method

3.3.1 Conceptual Designs

1) Low Pass Filtering

The low pass filter stage has been designed to allow frequencies to pass up to a corner frequency of 1000Hz. Beyond this, the signal is attenuated causing high frequencies to be blocked. This is important because the frequency range of alien pulse is between 135-1000Hz and there are external objects such as rocks in the environment which emit pulses outside of this frequency range. The trade-off from filtering out the high-frequency components is a reduction in the signal amplitude and smoothing of the signal edges which causes a loss of accuracy [1]. The amplifier circuit designed (figure 3.3) is a two-stage Low Pass Butterworth filter using Analog Filter Wizard and the simulated magnitude plot is shown in Figure 3.4.

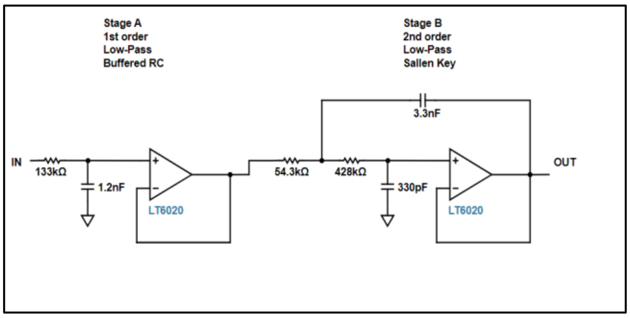

Figure 3.3: 2-Stage Butterworth Low Pass Filter

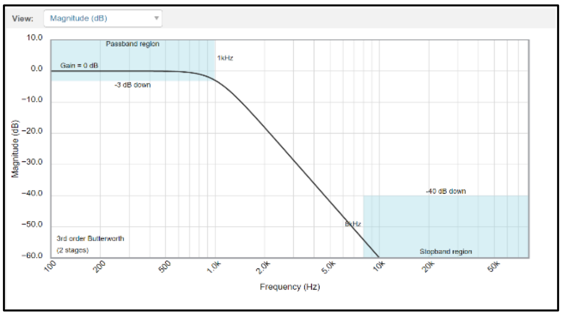

Figure 3.4: Magnitude Plot [4]

Simulating the low pass filter (figure 3.5) on LTSpice provides the expected characteristics of a signal with a reduced amplitude as seen from the blue output trace in figure 3.6. The reduction in amplitude can be increased using an amplifier circuit. Furthermore, a 2.5V DC-Offset has been added to ensure that the negative parts of the graph are not removed from clipping. Coupling capacitors are included at the input to prevent the DC-signal from being amplified as the source DC offset is unknown [16]. Coupling capacitors are added at the output to remove the DC offset.

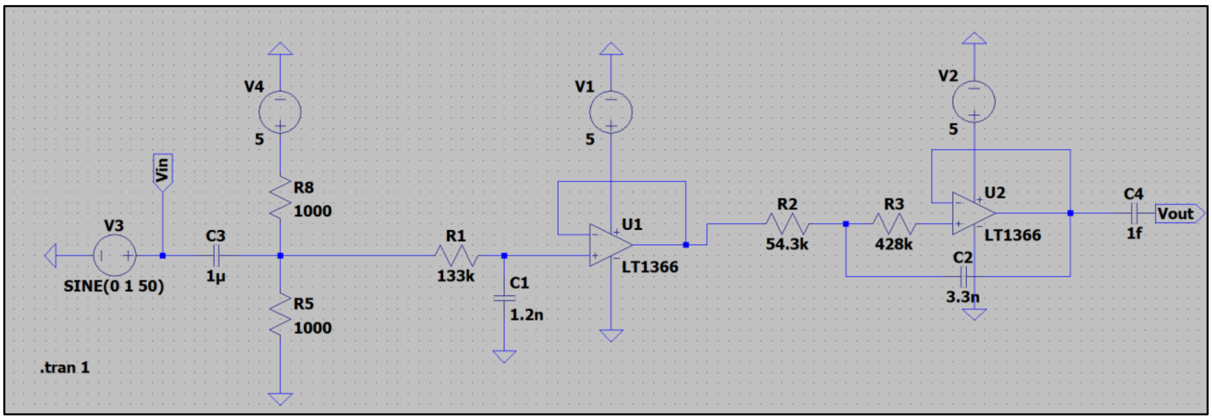

Figure 3.5: Low Pass Filter LTSpice Schematic

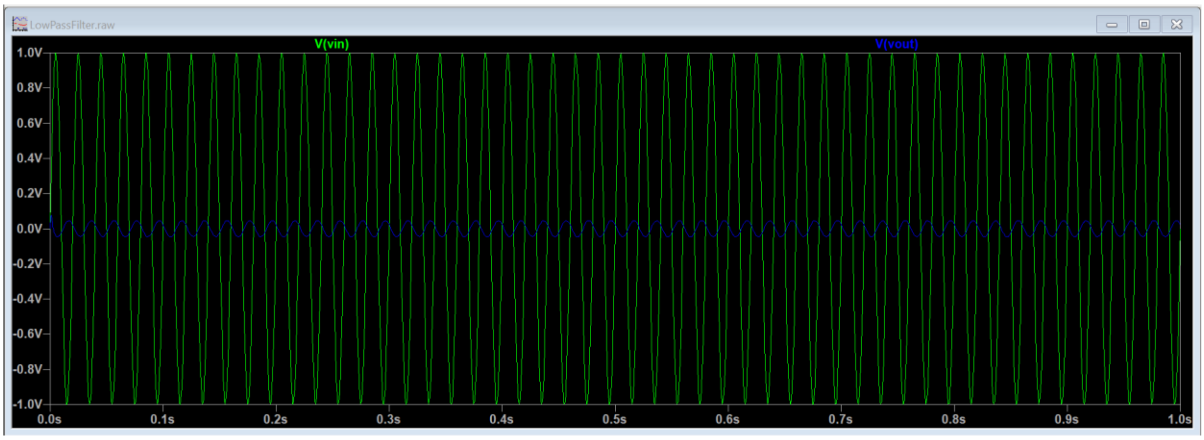

Figure 3.6: Transient Analysis (Low Pass Filter)

2) Amplifier Circuits

Inverting Amplifier

Initially, an inverting amplifier was designed to ensure the output signal was large enough to be detected. However, when simulated it was observed that the three negative peaks of the waveform were being amplified – figure 3.7. This is a problem because these peaks represent one pulse which makes it difficult to distinguish between successive pulses. Therefore, the suggested solution was to try a non-inverting amplifier and consider other methods to reduce the smaller ripples.

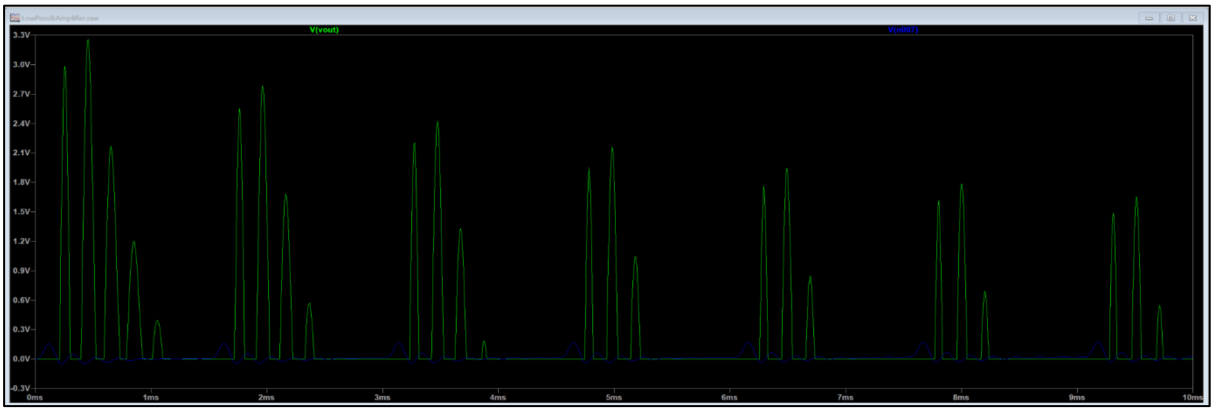

Figure 3.7: Signal passed through an Inverting Amplifier

Non-Inverting Amplifier

Implementing a non-inverting amplifier amplifies the highest positive peak of the blue trace in Figure 3.6, which results in a series of distinguishable peaks. This gives the desired outcome – Figure 3.8.

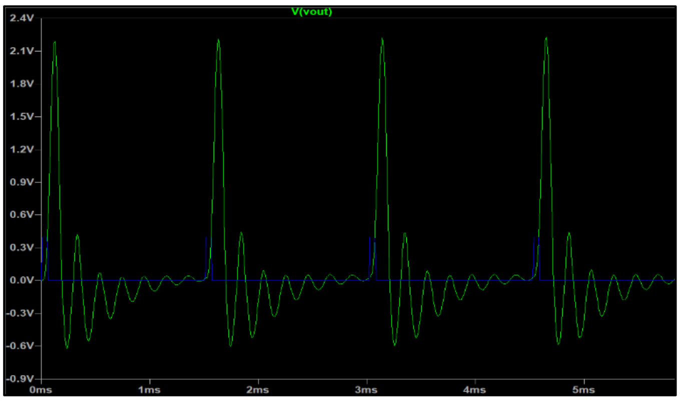

Figure 3.8: Non-Inverting Amplifier Waveform

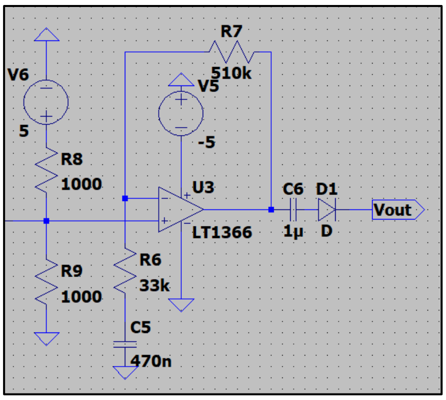

Figure 3.9: Non-Inverting Amplifier

3) Programming to determine the Period

After passing the pulse signal through the previous stages, the amplitude of the pulse was 1.5-2V when built and tested on a breadboard. The generated voltage signal outputted from the amplification stage was fed to the analogue signal pin on the microcontroller and programmed to calculate the time difference between the peaks. The initial approach was to detect the number of rising signals in three seconds and record the time difference from the total number. Then divide this time by the total number of pulses recorded in that interval to calculate the period. The analogue pin on the microcontroller was not suitable because it took too long for the analogue signal to convert to a digital signal. This resulted in an overlap between the detection time of each time difference which led to a loss of signal.

A suggested solution was to feed the signal directly to a digital input pin instead of the analogue pin. However, the amplitude of the signal was not high enough to be used directly in the digital inputs. Therefore, a device such as a Schmitt trigger would be useful because it could amplify the signal and perfect signal shape.

4) Schmitt Trigger

An Inverting Schmitt trigger was built using the LT1366 OpAmp with a positive supply voltage of 5V. A Schmitt trigger (comparator) works by setting a threshold value [17]. This was determined by looking at the positive peak of the analogue waveform and observing a value which is consistent with successive peaks. A voltage divider was designed at the positive interface of the OpAmp to provide a 1.3V DC reference voltage which acts as a threshold. This means that when the signal received becomes higher than 1.3V, the voltage will drop to approximately 0V acting as a flip-flop. Therefore, the output of the Schmitt trigger with no input theoretically maintains a continuous 5V signal. This design results in a series of pulses each time the infrared pulse is received. The blue traces from the LTSpice experimental simulations shown in Figure 3.10 display that the signal drop lasts around 20µs. The time interval between two adjacent signal drops is 5.3ms. As a result, there is sufficient time for the signal to be distinguished between being a high-value and a low-value voltage which can be fed into the digital signal interface of the microcontroller.

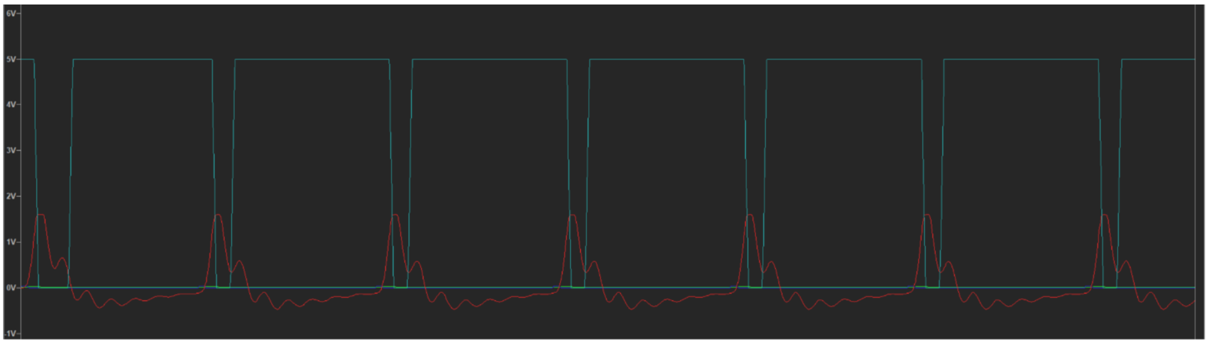

Figure 3.10: Red: Analogue signal output from non-inverting amplifier / Blue: Digital Signal Output

An inverting Schmitt trigger using an OpAmp circuit was chosen over the component itself because the time taken to order was too long and would further add to budget costs. Furthermore, the team had the components required to build the circuit from scratch using the LT1366 OpAmp. Large resistors were added to the inputs of the negative OpAmp input (see Figure 3.11) to draw away any excess current, which ensured that ideally no current flows into the input terminals of the OpAmp.

A major problem was the sensitivity of the infrared sensor circuit to picking up the input signal. To improve, the threshold value of the comparator was reduced to a smaller value by changing the resistor value ratio. This increased the range over which the infrared sensor worked – see section 3.4.1.

5) Final Program

In the initial test program, the number of signals received in a time interval of 500ms was averaged to determine the exact time for each signal. However, consideration had to be taken for the presence of measurement errors or a loss of signal due to a change in the position of the threshold set. Therefore, a new program was written that detected a LOW signal and stored the time in microseconds. It waited for another low signal and stored the time again. The time difference was calculated which allowed the period to be derived directly – full code found in appendix 6.

3.3.2 Final Design

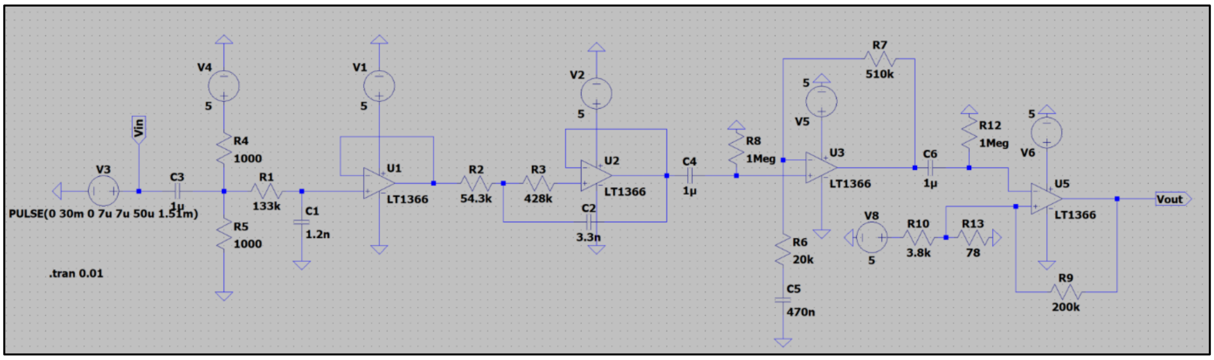

Figure 3.11: Final Circuit Design including Low Pass Filter, Non-inverting Amplifier and Schmitt Trigger

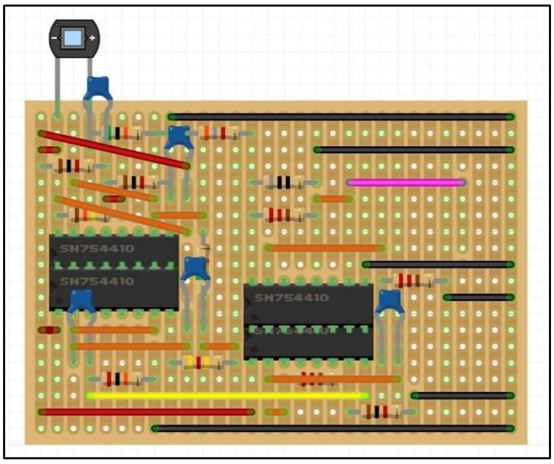

Figure 3.12: Fritzing Software – Designed Stripboard Schematic

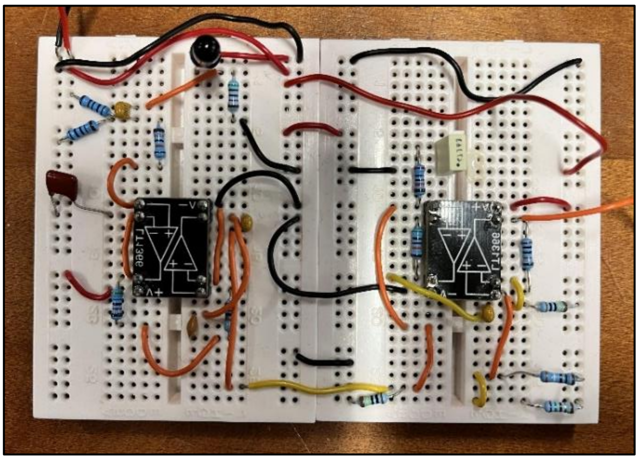

Figure 3.13: Prototype Breadboard Schematic

Figure 3.11 shows the final circuit design including each stage on LTSpice simulation software. This has then been designed on Fritzing (Figure 3.12) to implement the prototype circuit produced on the breadboard (Figure 3.13) onto a single-sided stripboard which provides a more compact and lightweight circuit.

3.4 Testing and Results

3.4.1 Range Test

Aim: To determine the operating range (distance) of the Infrared Detector.

Method:

1) Connect the oscilloscope probe to the output port of the age detection circuit and ground the ground probe.
2) Place the infrared diode at an initial range of 0cm from the alien and record the age of the alien by determining the period of the pulse.
3) Increase the distance between the infrared diode and the infrared emitter from the alien by 1cm. Record the age of the alien.
4) Repeat, increasing the distance by increments of 1cm until the signal is no longer detected to determine the maximum distance for valid detection.

Results:

Distance (cm)	Voltage: Peak-to-Peak (V)
0	5.12
1	5.28
2	5.32
3	5.32
4	5.16
5	5.16
6	5.31
7	5.28
8	5.16
9	5.16
10	4.44
11	0.24

Table 3.3: Range Test

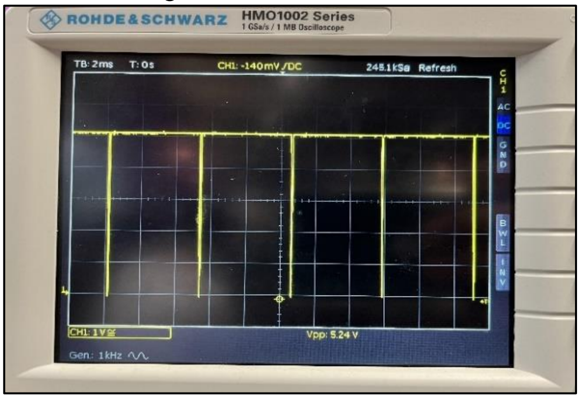

Figure 3.12: Digital Output Waveform

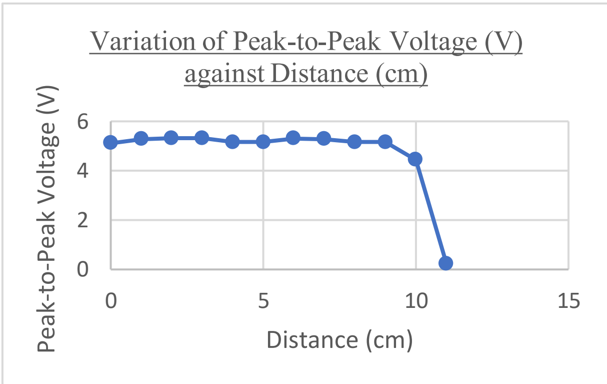

Figure 3.13: Variation of Voltage against Distance (cm)

3.4.2 Testing different Alien ages

Aim: To determine the age of different aliens and rocks.

Alien	Age (Year)
Red Alien	150
Purple Alien	584
Orange Alien	170
Blue Alien	682
Rock	n/a

Table 3.4: Testing aliens

3.5 Conclusion and Evaluation

From the first experiment, the range of the age detection circuit varies between 9-10cm in distance. This is supported by Table 3.3 where after 10cm, the voltage peak-to-peak is no longer equal to 5V, indicating a falloff in the intensity of the infrared signal which no longer exceeds the threshold value set within the inverting Schmitt trigger. Although the range is larger than predicted in the design specification, to avoid detecting the wrong alien, the rover must get close to the alien (within 3cm). Whilst testing, the focus of the photodiode was observed as being small, causing a drop in the signal when not correctly aligned. Hence, within the design of the chassis, the sensor was placed at a height equal to the alien's mouth which is approximately 43mm.

The second test indicated that the circuit and program correctly identified the age of different aliens, but no age was recorded for the rock (table 3.4). This is because the frequency emitted by the rock is outside of the range of the low pass filter (135-1000Hz) [1]. This avoids the chance of detecting the wrong signal which satisfies the design specification of only detecting infrared signals.

Evaluating the design specification (table 3.2), all the criteria have been appropriately met including designing a compact circuit that fits on a third of the stripboard, allowing space for the other sensor circuits. A problem faced when soldering all the components onto a stripboard was the presence of excess noise. Due to time constraints, it was decided that the breadboard prototype will be used in the final rover on demo day.

4. Magnetic Field Detection

4.1 Theory

4.1.1 Aims

The aliens present in the arena possess a magnet in their body, shown in Figure 1.3, which results in a magnetic field being created around the alien. According to the specifications of the rover [1], it must be detected whether this field is negative, positive or neutral.

4.1.2 Choice of Sensor

To detect the magnetic field, several sensors could be selected, as outlined in Table 4.1.

Method	*Advantage*	*Disadvantage*
Magnetometer	• Outputs a very precise measurement of the Earth's magnetic field • Built-in filtering with little noise	• Difficult to measure the true impact of the alien on the measured magnetic field (the Earth's) • Expensive (£3.5 each), no contingency
Digital Hall Effect Sensor	• Detects the presence of the magnet at a high precision • Unaffected by the Earth's magnetic field • Cheap • Low current consumption	• Does not differentiate between positive and negative field • Difficult to calibrate • Range is unknown
Linear Hall Effect Sensor	• Detects the presence of the magnet at a high precision • Unaffected by the Earth's magnetic field • Built-in filtering • Cheap • Outputs a voltage proportional to the magnitude of the magnetic field, in the same direction	• Range is unknown • Sensitive to noise

Table 4.1: Magnetic Field Sensors

From Table 4.1 a linear Hall Effect sensor was the best choice, as it is low cost and can detect the direction of the magnetic field, while the other options cannot. Testing then took place to determine if it met the other specifications highlighted in the next section. Many linear Hall Effect sensors exist, differing only in their sensitivity due to different choices of integrated circuits and semiconductor choices.

A Hall effect sensor consists of a simple flat semiconductor plate, through which a small current flows, as seen in Figure 4.1. When no magnetic field is present, the charge carriers flow uniformly across the plate. However, when a magnetic field is applied, the charge carriers will move perpendicular to the current flow. Positive charges move in the opposite direction of negative charges, thus creating a voltage across the plate [18]. The directions change when the magnetic field direction changes, thus the voltage would be positive or negative depending on whether the magnetic field is positive or negative. The magnitude also changes, but this is not necessary for the specification of this project, considering the measured magnitude of the magnetic field is dependent on the distance to the sensor, so holding the sensor further away from the magnet would decrease the Hall voltage output.

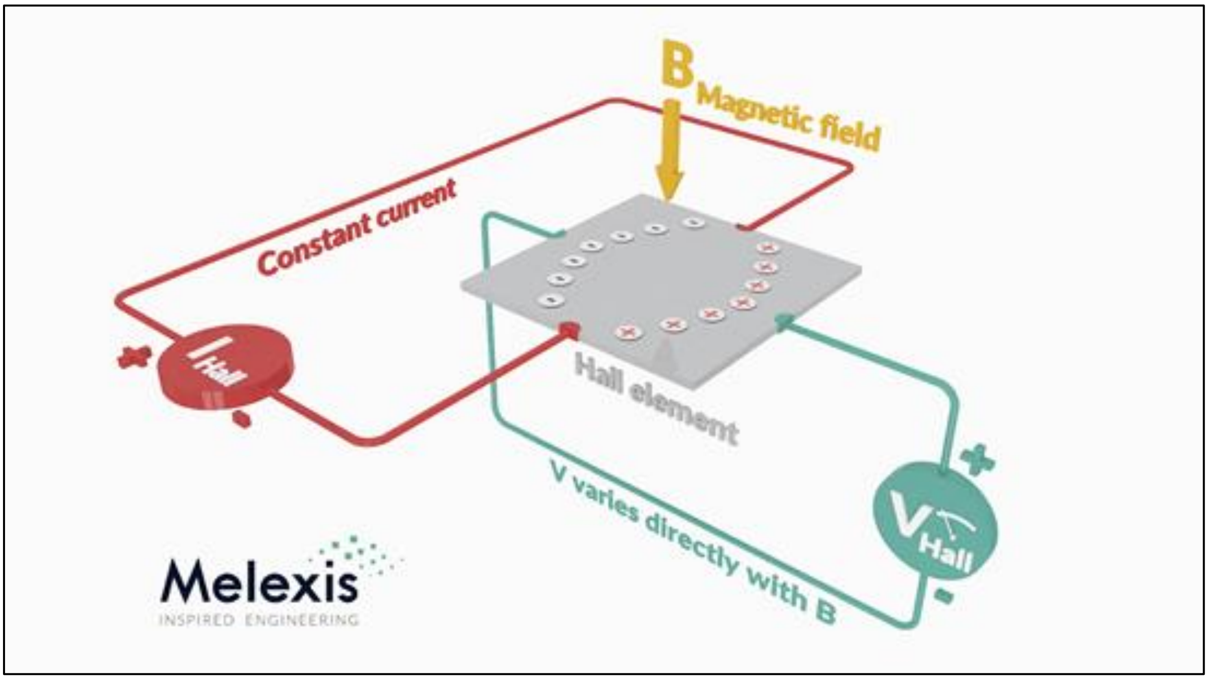

Figure 4.1: Hall effort sensor workings [36]

Therefore, the Honeywell Linear Hall Effect Sensor SS49E was selected [19], with the reasons for this choice outlined in Table 4.2.

Specification	Quantitative value	Reason for optimality
Price (<£5)	£0.68 each	Multiple sensors can be bought in this budget, to ensure no defects in manufacturing would impact the project.
Range	Unknown	The range for the sensor is unknown, testing will need to be carried out.
Temperature (15˚C to 30˚C)	-40˚C to 100˚C	The sensor's operating temperature range far exceeds the required operating range.
Operating voltage	2.7V to 6V	The Adafruit and Orangepip boards both have 5V out pins, which are within this range.
Current consumption (<8mA at 5V)	6mA at 5V	This value given is within the range. Testing will be carried out to ensure this value from the datasheet is accurate.
Stable output	Built-in filtering	This foregoes the need to build a filtering circuit, thus decreasing costs and space requirements.
Delivery time (1-2 business days)	2 business days	This short delivery time ensures waiting for the sensor to arrive does not delay the project too much, and that the time requirements are met.

Table 4.2: Hall effort sensor workings [36]

4.2 Design Specification

Specifications required for the magnetic field detection, are outlined in Table 4.3

Feature	Quantitative specification	Qualitative specification
Must detect the direction of an alien's magnetic field.	Output -1, 0 or 1, corresponding to a negative, neutral or positive magnetic field.	This is given as part of the design brief for this project. These output values are chosen for readability.
Must have sufficient range.	>5cm	Due to size and manoeuvrability constraints, the range must be as large as possible.
Must be lightweight.	<10g	The sensor must not affect the mass of the rover, since it must be lightweight overall
Must be low cost.	<£5	The rover must be cost-effective as given in the design brief. The sensor should not cost more than the allocated budget.
Must be energy efficient.	<8mA at 5V	Low energy consumption ensures more power can be used for the motors, increasing the overall speed of the rover, without needing heavier batteries.
Must work at a large temperature range.	15˚C to 30˚C	This ensures the robustness of the circuit, in a lab with varying temperatures.

Table 4.3: Magnetic Field Detection Design Specification

4.3 Experimental Method

4.3.1 Conceptual Design of the Circuit

Due to the built-in filtering, initial testing was conducted by connecting the sensor directly to the OrangePip microcontroller, as seen in Figure 4.2. From here, initial testing was conducted, for each of the 5 sensors that were ordered, to determine their range, and which sensor to use in the final circuit design.

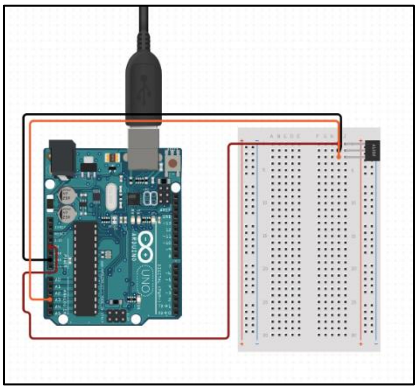

Figure 4.2: Microcontroller

4.4 Testing

4.4.1 Initial Range Test

Aim

To determine the range of each of the labelled 5 sensors (see Method 1) and collect quantitative data on their noise stability (see Method 2).

Method 1

1. Label each sensor from 1 to 5 using masking tape.
2. Connect sensor 1 to the OrangePip board as shown in Figure 4.2. Ensure the leftmost pin is connected to 5V, the middle pin to ground and the rightmost pin to an analogue in pin (A0 in this case).
3. Set up a ruler measuring distance away from the flat side of the sensor.
4. Run the C++ code (Figure 4.3) using the Arduino IDE on the OrangePip board.
5. Bring the magnet slowly closer and closer to the sensor, along the axis of the ruler.
    - Record the distance at which the output changes by ±10.
6. Repeat steps 2 to 5 for each of the 5 sensors.

Method 2

1. Repeat steps 1 and 2 from Method 1
2. Make sure the magnet is far from the sensor. Run the C++ code (Figure 4) using the Arduino IDE on the OrangePip board.
3. Record the output in a table.

```
void setup() {
  pinMode(A0, INPUT);
}

void loop() {
  Serial.println(analogRead(A0));
}
```

Figure 4.3: Arduino IDE (Method 1)

```
void setup() {
  pinMode(A0, INPUT);
  int output=0;
  for(int i=0;i<200;i++{
    output=output+analogRead(A0);
  }
  output=ouput/200;
  Serial.println(output);
}
void loop() {
}
```

Figure 4.4: Arduino IDE (Method 2)

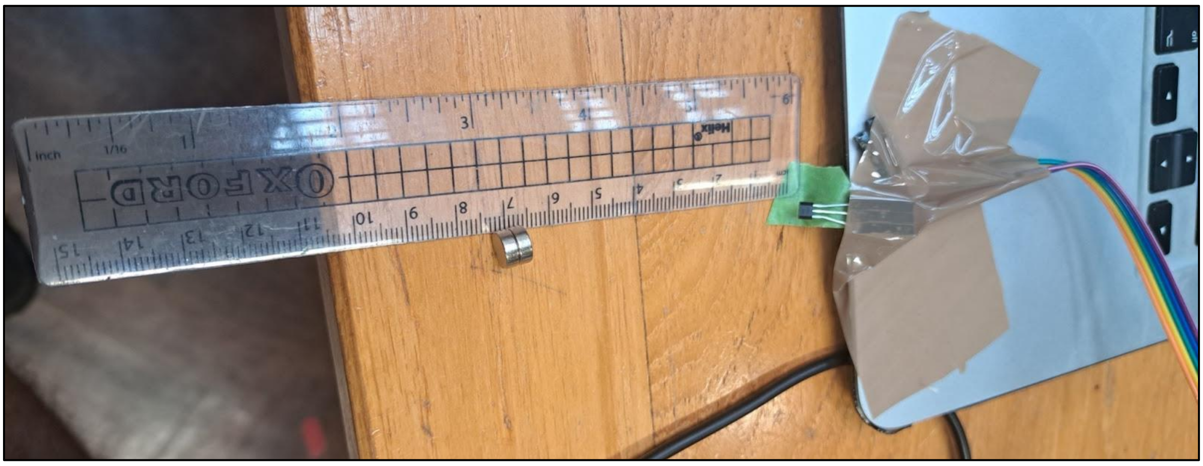

Figure 4.5: Set up of experiment in Method 1

Results

The sensor outputs a value between 0 and 1023 in the analogue pin A0. This corresponds to a voltage out value scaled up between 0 and 5V, where a middle value of 2.5V (or 511) corresponds to no magnetic field present. Column 2 contains the results of Method 2, and columns 3 and 4 contain the results from Method 1. These can be seen in Table 4.4:

Sensor	Output average given over 200 iterations	Distance at which output changes by more than 10 in the positive direction (cm)	Distance at which output changes by more than 10 in the negative direction (cm)
1	486	1.6	1.7
2	506	2.3	2.2
3	509	2.2	2.2
4	492	2.1	2.2
5	499	2.2	2.1

Table 4.4: Results from Method 1 and 2

Sensor number 3 is the most accurate sensor since its average output was the closest to 511, the target value when there is no magnetic field present. This sensor also has the same range for positive and negative magnetic fields. However, the range of 2.2cm is not satisfactory, as it does not meet the >5cm specification.

This 2.2cm range was measured with output changes of more than 10, which happens when the magnet is very close to the sensor. This testing, however, showed that the sensor detects the magnet at a much larger range, changing its output only slightly. Noise makes these small changes difficult to identify, forcing a smaller range to have satisfactory certainty.

Therefore, to improve the range the noise must be removed. Several methods were explored for this, as shown in Table 4.5:

Method	*Advantage*	*Disadvantage*
Filtering and amplification	• Can very precisely filter noise • Amplify small changes for more sensitivity	• Expensive and time-consuming • Low flexibility o Filtering hard wired • Takes up more space
Averaging inputs	• Simple and time efficient • Can be scaled up or down depending on needs	• Possibly a negligible effect if the noise is too high • Takes longer to execute

Table 4.5: Methods for removing noise

Since the averaging inputs method was the simplest, fastest, and cheapest to implement, it was chosen as a preliminary method of stabilising the noise. A C++ function was written to average analogRead() inputs over a certain number of readings. This way, random increases and decreases in the voltage should cancel, and a true, noise-stabilised value should be reached, which will be tested in the next section.

4.4.2 Noise Stabilisation Test

Aim

- To determine whether taking an average outputs satisfactory noise stabilisation and how many readings should be averaged together for optimal stabilisation.

- To determine the effectiveness of the stabilisation, a precision benchmark will be used. The closer the output averages are to each other, the more effective the stabilisation is.

Precision can be calculated by simply finding the standard deviation of the dataset, according to the formula in Equation 1:

$$Equation\ 1\text{:}\ \sigma = \sqrt{\frac{\Sigma(x-\mu)^2}{n-1}}$$

The lower the standard deviation, the more the noise has been removed.

Method 3

1. Repeat steps 1 and 2 in Method 1 in the previous section.
2. Ensure the sensor is as far away from any magnetic interference as possible.
3. Run the C++ code std-dev-test (Figure 4.6).
4. Record the execution time and standard deviation for averages of 1, 5, 10, 25, 50, 100, 500, and 1000 readings.

```
// Outputs the averaged readings
float get_raw_field() {
  float output=0;
  for(int i=0;i<1000;i++){
    output=output+analogRead(A0);
    }
  return output/1000;
}
// Finds the standard deviation from the given input array
double std_dev(double values[100], int n) {
    double sum = 0;
    for (int i = 0; i < n; i++){
      sum=sum+values[i];
    }
    double mean = (double)sum/(double)n;
    double square_difference = 0;
    for (int i = 0; i < n; i++) {
      square_difference=square_difference+((values[i]-mean)*(values[i]-mean));
    }
    double variance=(float)square_difference/n;
    return sqrt(variance);
}
void setup() {
  pinMode(A0, INPUT);
  Serial.begin(9600);
  // Finds the time it takes to get raw results
  float start = micros();
  get_raw_field();
  float end = micros();
  Serial.print("The execution time was ");Serial.print(end-start);Serial.print(" microseconds.");
  Serial.println("");
  // Finds the standard deviation of 500 raw values
  double values[100];
  for(int i=0;i<100;i++){
    values[i]=get_raw_field();
  }
  double standard_deviation=std_dev(values,100);
  Serial.print("The precision was ");Serial.print(standard_deviation);
  Serial.println("");
}
void loop() {
}
```

Figure 4.6: C++ Code std-dev-test

Results

After conducting this test, using 100 averages to find the standard deviation, the results can be seen in Table 4.6:

Number of readings per average	Execution time (s)	Standard deviation
1	212	0.5
5	736	0.16
10	1336	0.11
25	3136	0.07
50	6140	0.06
100	12144	0.03
500	60324	0.02
1000	120568	0.02

Table 4.6: Results from Method 3

From this a clear correlation can be observed: increasing the number of readings per average decreases the standard deviation but increases the execution time. This linear behaviour can be represented by the formula in Equation 2:

$$\text{Equation 2: } y = 120.438x + 120.59$$

Where y is the execution time and x is the number of iterations.

One requirement is that the magnetic field should be detected in less than 10ms so that it does not block the other code of the rover. Solving Equation 2 with $y = 10ms$, gives the number of iterations which should be 82, rounded down to 80. Using Method 3 to test, the execution time was 9.74ms for a precision of 0.02, satisfying all conditions, with good noise stabilisation. This precision value also suggests no further precision gain with more iterations.

4.4.3 Development of the Code

To find when a magnetic field is present and its direction, the value that the sensor outputs when no magnetic field is present must be known so that when the value differs from this, the correct magnetic field direction can be identified. There are many ways to do this, as outlined in Table 4.7.

Method	*Advantage*	*Disadvantage*
Taking a range of possible values at the beginning of execution	• More robust, encompasses the range of values the sensor takes. • Longer calibration means more possible values accounted for	• Long calibration time • If too much noise decreases the range
Manually entering a range of values once	• Simple to do; long calibration can be performed. • More possible values accounted for	• The sensor sensitivity drifts, what is a none value now could be a magnetic field tomorrow

Table 4.7: Calibration Methods

From this, taking a range of possible values (lowest and highest output by the sensor over the calibration period) at the beginning of the execution of the code is the most robust solution. While this may decrease the range, it offers the advantage of being highly adaptable and less sensitive to noise.

A simple calibration function was developed as seen in Figure 4.7. When called, this function iterates for a given number of stabilised readings, recording the highest reading and lowest reading over this period, and outputs this range. When a reading is within the range, there is no magnetic field.

```
// Calibrates the Hall Effect sensor
void calibrate(){
  float min_value=1023;
  float max_value=0;
  for(int i=0;i<100;i++){
    float magnetic_field=get_raw_field();
    if(magnetic_field <min_value){
      min_value=magnetic_field;
    }
    if(magnetic_field>max_value){
      max_value=magnetic_field;
    }
  }
  min_max[0]=min_value;
  min_max[1]=max_value;
}
```

Figure 4.7: Calibration Program

4.4.4 Test of the calibration function

Aim

To determine whether the calibration function increases the range, the execution time of such a function, and the optimal number of iterations in calibration.

Method 4

1. Repeat steps 1 and 2 in Method 1 in the previous section
2. Ensure the sensor is as far away from any interference as possible
3. Run the C++ code, named calibration_test.ino and record the results
    - Repeat this 3 times and take the average
4. Record the range using the same setup as in Method 1
5. Repeat steps 2, 3, and 4 for 100, 250, 500, 1000, 2000 and 5000 iterations in the calibrate() function

Results

The results of the experiment can be seen in Table 4.8 below:

Number of iterations in calibrations	Range of sensor (cm)	Time to execute (ms)	False positives per second (average over 3 trials)
100	13.0	605	1.67
250	12.5	1512	0.55
500	12.5	3024	0.43
1000	12.0	6050	0.18
2000	12.0	12101	0.14
5000	12.0	30254	0.02

Table 4.8: Results from Method 4

It is clear from these results that longer calibration periods result in a slight decrease in range and a decrease in the number of false positives each second (when there is no magnetic field). However, the execution time of the function increases linearly, and as such, the best tradeoff between calibration time and the number of false positives must be found. To investigate this tradeoff, the data was plotted in Figure 4.8.

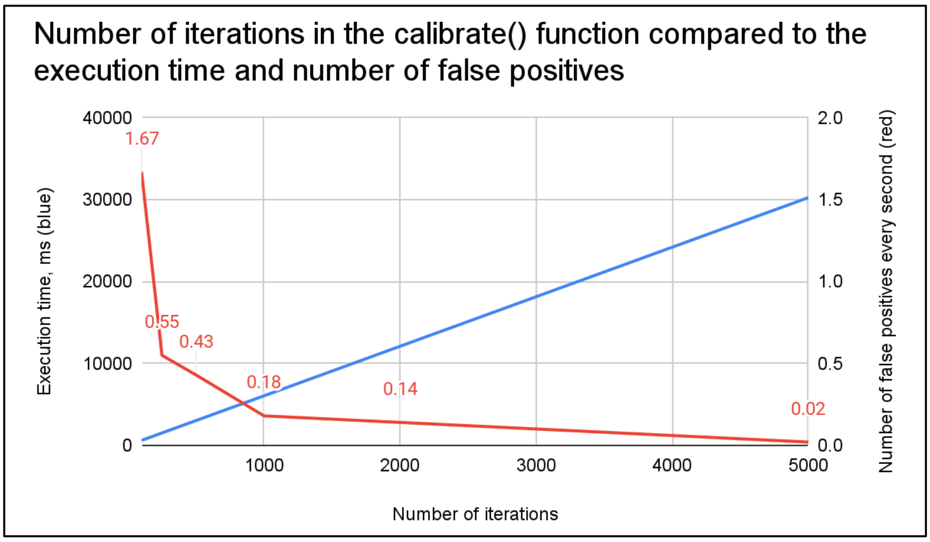

Figure 4.8: Plot for calibration function

As the rover may have to recalibrate occasionally, the calibrate() function cannot take forever. Therefore, 500 iterations will be used, as any reduction in false positives after this will be minimal.

4.5 Conclusion and Evaluation

The testing aspects of the calibrate() function were then removed, and the code was organised in a class, as seen in the final code here.

Further testing was then conducted against all the specifications, shown in Table 4.9.

Specification	Experimentally measured value	Notes	Passes test?
Must output -1,0,1	The system was able to output -1, 0, and 1 as specified.	Passed	Yes
Price (<£5)	Individually: £0.68 Total (x5): £3.4	5 sensors were bought for a price well below the budget,	Yes
Range (>5cm)	12.0cm	The range of the system is more than twice the minimum range, allowing for great system flexibility.	Yes
Temperature (15˚C to 30˚C)	15˚C to 30˚	The system was tested at various temperatures in the lab.	Yes
Current consumption (<8mA at 5V)	6mA at 5V	Passed	Yes
Mass of the sensor (<10g)	4.3g	Passed	Yes

Table 4.9: Final design meets all specifications

Subsequently, the magnetic field detection system designed in this section passes all required tests, and as such, can be integrated with the other submodules, to detect the magnetic field of the alien.

5. Remote Controlling and Data Output (I/O)

5.1 Theory

5.1.1 Aims

This section aims to allow user input and remotely control the rover, as well as communicating alien data back to the user. It addresses two requirements given by the criteria:

1. The rover must be manoeuvrable enough to negotiate the environment.
2. The remote-control interface must be logical and easy to use.

5.1.2 Remote Controlling-Input

To create a remotely controlled rover that can explore a remote planet, the rover must have a human input method. As it will have continuous real-time communication with the rover, the bot does not need to be autonomous, but even so, there must be a way for a human to interact with the rover.

With consideration of the final specification (Appendix 3), a phone/controller method works best for the EEERover. Due to the time constraints, it would be extremely difficult to develop a functional mobile app within the timespan set by the client. Due to this, and the fact that a controller would allow a greater variety of actions, a controller input system was chosen to be developed. A suitable alternative that will be developed as a fallback option is a keyboard input system. Table 5.1 highlights the advantages and disadvantages of alternative input systems that were considered to control the rover.

Method	*Advantages*	*Disadvantages*
Keyboard	• Compatible with most devices • Easy to implement	• Lack of precision • Controls unintuitive
Phone	• Intuitive controls • Widely accessible • Precise movements possible	• Difficult to develop, distribute and download
Controller	• Intuitive controls • Precise movements possible • Easy to control	• Requires a dedicated controller and an app
Autonomous	• Does not require human input	• Unreliable

Table 5.1: Alternative input systems

5.1.3 Remote Controlling-Communication

Data needs to be communicated between the user and the rover to display the characteristics of each alien encountered and to control the rover.

Since the rover is required to be cost-effective, it would be a waste not to use the given EEEBug expansion kit, which includes a Wi-Fi shield. For this reason, Bluetooth will not be used but it is still a valid solution.

There exists a variety of protocols for transporting data. The OSI (Open Systems Interconnections) model describes the layers, in which the lowest layer describes transporting individual bits over a physical medium and the highest layer describes high-level protocols for resource sharing [20]. As everything else is taken care of either implicitly or by another application, only the 4th and 6th layer are included in this investigation - see Figure 5.1.

OSI model				
Layer			Protocol data unit (PDU)	Function[26]
Host layers	7	Application	Data	High-level protocols such as for resource sharing or remote file access, e.g. HTTP.
	6	Presentation		Translation of data between a networking service and an application; including character encoding, data compression and encryption/decryption
	5	Session		Managing communication sessions, i.e., continuous exchange of information in the form of multiple back-and-forth transmissions between two nodes
	4	Transport	Segment, Datagram	Reliable transmission of data segments between points on a network, including segmentation, acknowledgement and multiplexing
Media layers	3	Network	Packet	Structuring and managing a multi-node network, including addressing, routing and traffic control
	2	Data link	Frame	Transmission of data frames between two nodes connected by a physical layer
	1	Physical	Bit, Symbol	Transmission and reception of raw bit streams over a physical medium

Figure 5.1: OSI model representation [20]

There are 2 commonly recognised ways to send data on the 4th (i.e., transport) layer, as referenced in the model. Table 5.2 indicates a comparison of these two methods.

Method	*Advantages*	*Disadvantages*
TCP	• More reliable • Error-detection capabilities	• Connection time (requires 3 packets to set up socket) • Higher latency • Bigger header size (20 bytes)
UDP	• Connectionless (fast setup) • Low overhead (8 bytes)	• Unreliable (packets may be lost)

Table 5.2: Comparison of UDP and TCP [37]

For the rover, UDP (User Datagram Protocol) communication will allow for low latency as it does not require the additional data processing that TCP (Transmission Control Protocol) requires. As the data stream used will be continuous, a loss of data packets is not an issue. This is because multiple packets will be fired every second. Therefore, as per the specification, UDP has been chosen.

Additionally, the method of encoding data on UDP packets must be chosen – this refers to layer 6 of the OSI Model. Table 5.3 indicates different methods of transporting data and the reasons for choosing them.

Method	*Advantages*	*Disadvantages*
JSON	• Lightweight • More secure	• No support for complex data types
XML	Allows complex data types (such as binary data/timestamps) • Easy data validation	• More verbose • Parsing documents expensive computationally

Table 5.3: Alternative data transport methods [38]

JSON (JavaScript Object Notation) has been selected over XML (Extensive Markup Language) due to its lightweight nature.

5.1.4 Data Output

The method of data output is highly dependent on the choice of input and communication. For example, with a keyboard/TCP, a computer website using HTTP would be appropriate for displaying data.

Since the team proceeded with controller/UDP, a computer app would be used to display data sent back from the alien, as it allows for:

- Easy translation between controller input and UDP packets.
- Continuous output to be shown.

To develop an app that supports UDP and controller input, a platform needed choosing that allowed quick development. Table 5.4 indicates potential programming platforms that were considered.

Method	*Advantages*	*Disadvantages*
Java	• Scalable • Cross-platform compatibility	• Verbose
C#/.NET	• Integration with Windows • Scalable	• Platform limitations
Python	• Simplicity	• Performance

Table 5.4: Alternative development platforms [39]

As the only part that needed to be shown is the alien output, the time spent developing an app needed to be kept to a minimum. For this task, a simple Python interface was developed.

5.2 Design Specification

Table 5.5 identifies the attributes specific to the I/O section.

Feature	Quantitative specification	Qualitative specification
Performance		Continuous transfer of data for ease of use: no need to press button to collect data.
Low latency	Average latency >100ms	Must be responsive enough to be easily manoeuvred
Reliability	<200ms of latency within predicted operational time (4 mins)	Must be reliable enough to not disconnect at critical times (e.g., collecting data)
Simple GUI	No menus or buttons	Ease of use, better user experience.

Table 5.5: I/O Design Specification

5.3 Experimental Method – Implementation

5.3.1 Planning Ahead: enabling collaboration and setting conventions

This section involves developing the core of the final program. Therefore, to enable collaboration and easier integration, a GitHub was created. The README.md file (Figure 5.2) was set up with instructions on using GitHub and an internal guide for developing code into the database, allowing the rest of the team to work easier. See Appendix 6 for the full GitHub code.

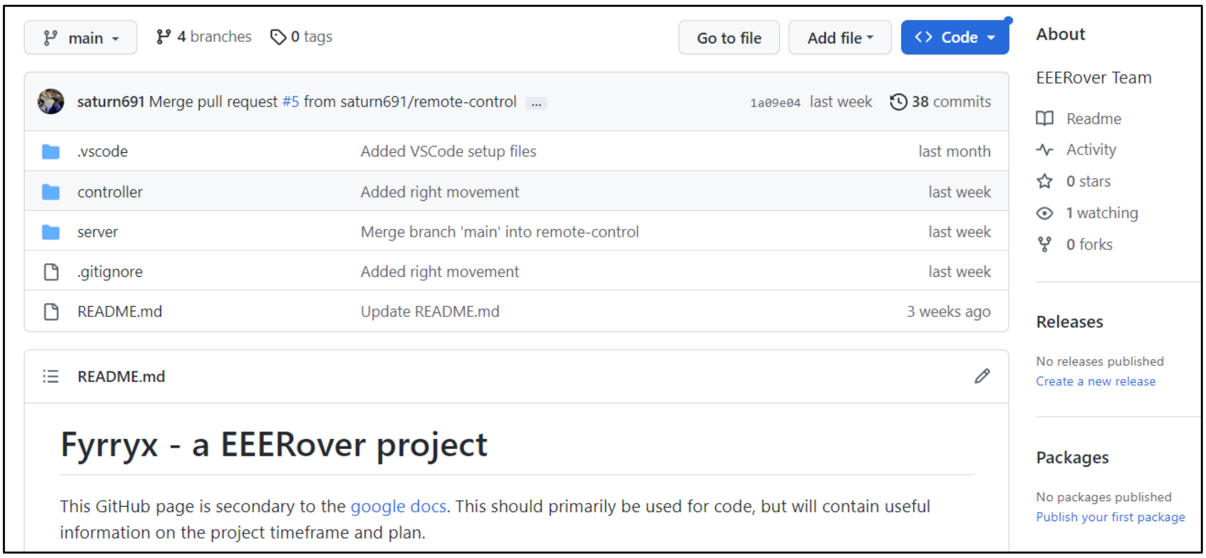

Figure 5.2: GitHub main page

One advantage of using git is for version control and branching. This allows for going back to a previous part of the code if necessary. GitHub was used to enable collaboration, with the main branch protected, only containing working code and any other code pushed into a subbranch before being merged into the main branch.

To maintain clean and readable code, standard industry practices were used. This includes:

- Using OOP (object-orientated programming) techniques.
- Following styling conventions such as PEP 8 [21].
- Using file decomposition.

5.3.2 Overview – Process

For both the controller/server, a setup and a loop are needed. In the setup process,

1. The server connects to a set IP address and port.
2. Controller pings set the IP address and port.
3. When the server receives a “ping”, it replies with a “pong”. This also fetches the latency (see section 5.4.1).
4. Controller receives the “pong”, is happy with the connection and starts the program.

In the loop process,

1. Controller sends movement data to the server.
2. Server handles the movement data.
3. Server sends signal data to the controller.
4. Controller displays signal data.
5. Controller handles any other user-requested inputs.

5.3.3 UML (Unified Modelling Language) Diagrams

The server and controller can be represented in a UML class diagram, which are in Figures 5.3 and 5.4 respectively. The diagrams show the system classes, their attributes, operations and relationship among objects [22].

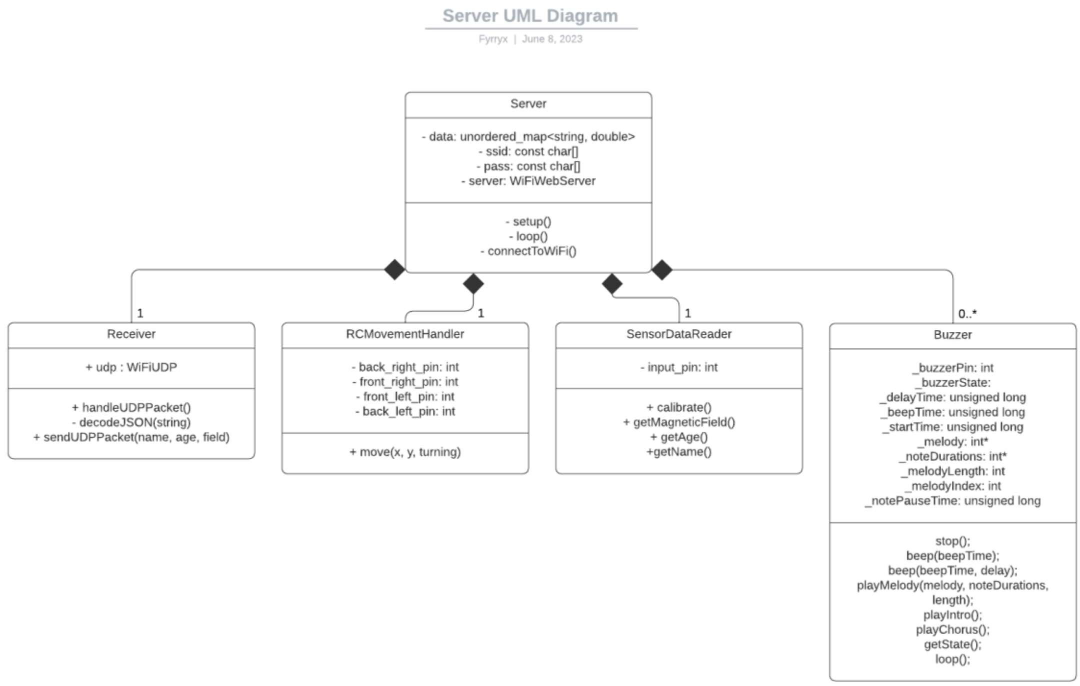

Figure 5.3: Sever UML Representation

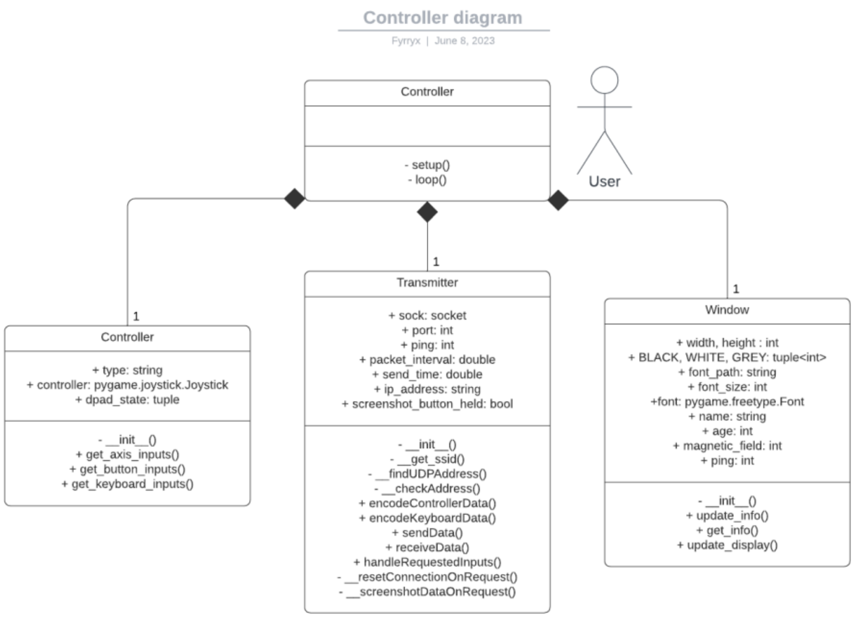

Figure 5.4: Controller UML Representation

5.4 Testing

5.4.1 Finding the latency of the controller

Aim:

To find the latency (ping) of the controller, when the controller sends the JSON message to the server, the server responds with a “pong” message, which is being listened to by the controller.

Figure 5.5 provides a code snippet from the sendData() function from the class Transmitter (see Figure 5.4: path - Transmitter).

```
132     # Sends data to the server to be processed and handles resetting
133     def sendData(self, data):
134         if self.send_time + self.packet_interval > time.time():
135             return
136 
137         json_data = json.dumps(data) # Packages the data into a JSON file
138         print(json_data)
139 
140         server_address = (self.ip_address, self.port)
141         self.sock.sendto(json_data.encode(), server_address) # Sends the data to the server
142         try:
143             data, addr = self.sock.recvfrom(1024) # Waits until a "pong" is received from the server
144         except:
145             self.ping = 9999 # There is a timeout of 100ms, so this activates when it timeouts
146             return
147 
148         end_time = time.time() # Ends the timer
149         ping = (end_time - self.send_time) * 1000 # Works out the difference in ms
150         self.ping = ping  # Allows the user to fetch the ping
151 
152         self.start_time = time.time() # Starts the timer
```

Figure 5.5: Path Controller > transmitter.py > Transmitter > sendData()

This function measures the total delay between a possible controller input (at the end of the function: line 152) and the time it takes for the whole program to run until it is called again in the loop, then waits until a “pong” is received from the server (line 143) before updating the value of ping (line 150) to be shown in the GUI.

5.4.2: Allowing precise movements

To measure the reliability, the ping can be graphed over iterations – figure 5.6. To do this, the transmitter.ping object (see figure 5.4: Transmitter) is written to a text file on every loop, and the program was run for double the predicted operational time (4 mins). Then, Excel was used to analyse the data.

From data analysis:

- Mean ping (includes anomalies): 8.808411ms
- Median ping: 5.105019ms
- Lag spikes (times in which ping exceeds 100ms): 13
- Longest time without lag spike: 177.5889s (between iteration 27883 and 56025)

On closer inspection, the lag spikes were a by-product of packet loss, as there were no consecutive lag spikes. This is due to the very nature of UDP and although the connection is still reliable, these spikes will effectively increase the ping of the next packet by 100ms.

Therefore, it can be concluded that the connection is reliable, and the average latency is low enough.

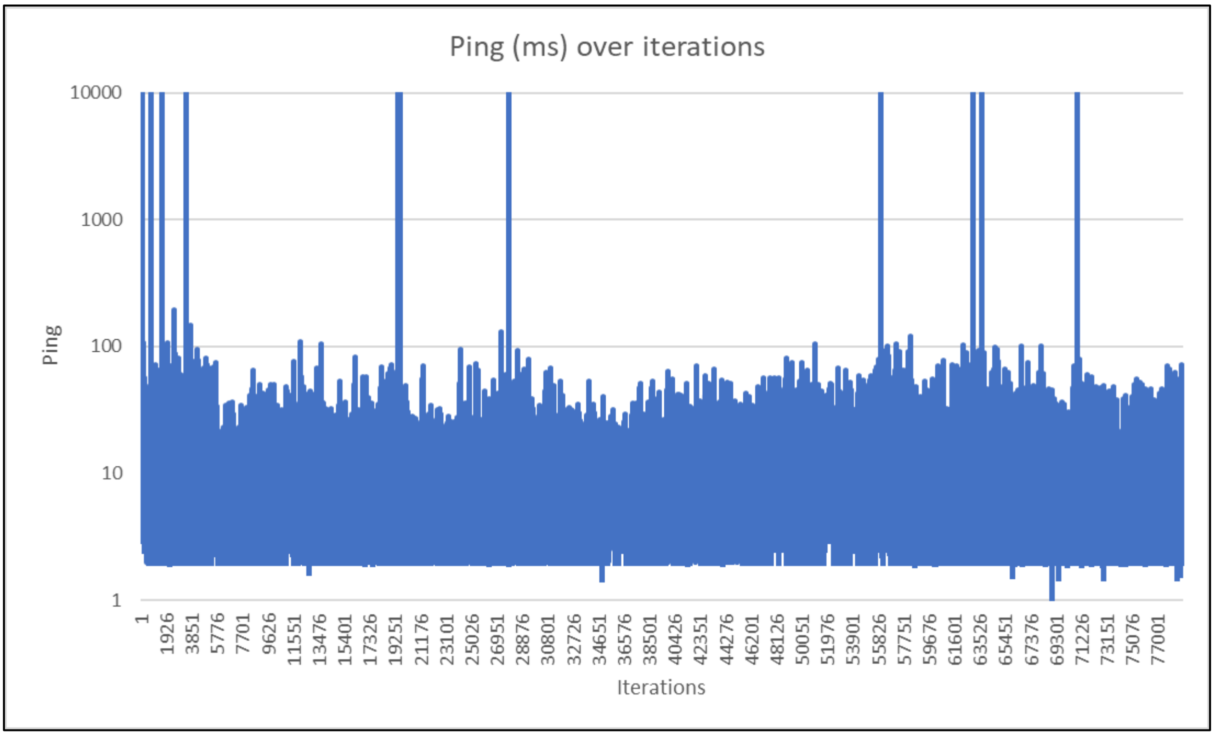

Figure 5.6: A logarithmic graph showing ping recorded per iteration of the controller loop

5.4.3 Creating a logical and easy-to-use interface

To address the specification of not having any buttons or menus, a simplistic, table-inspired interface (figure 5.7) was built using pygame 2.4.0 [23]. This supports controller and keyboard controls; therefore, no buttons or additional clutter is required.

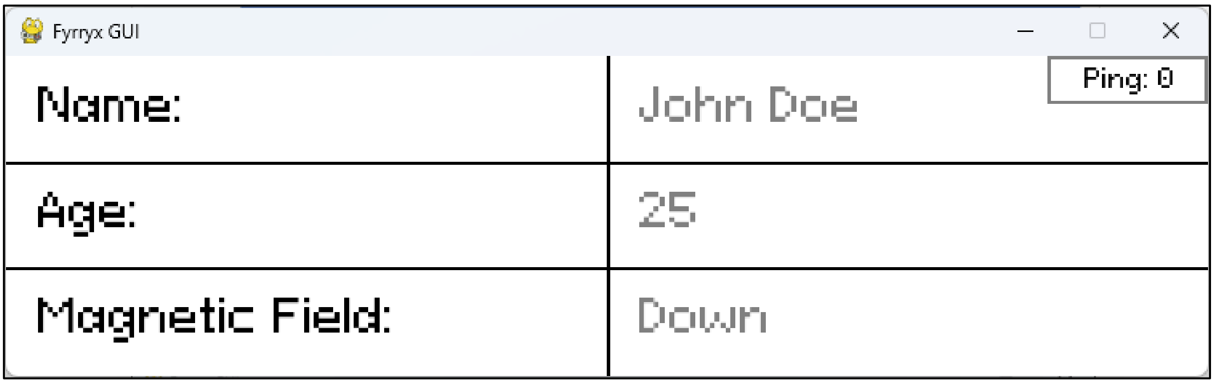

Figure 5.7: Data Output GUI

5.5 Evaluation

Looking at the specification, all points have been addressed and criteria met. This submodule could be further developed in other ways, including:

- Creating a mobile controller app, allowing easier connectivity.
- Limiting the number of devices able to connect to 1, preventing multiple devices from sending packets.
- Adding a security protocol on top, preventing possible (D)DoS attacks.

6. Design of Chassis and Motor Control

6.1 Theory

Aims

The EEERover must be designed and constructed that considers the budget and weight limits [1] whilst fitting the sensor circuits.

6.1.1 Wheel Type

The first stage for the design of the chassis is the wheel type for the rover. Multiple wheels were considered including:

- Omni Wheels – An omni wheel is a wheel that has rollers placed around the circumference of the wheel perpendicular to the rolling direction of the wheel (figure 6.1)
- Traditional Wheels - (figure 6.2).
- Mecanum Wheel - A wheel that has rollers placed around the circumference of the wheel 45 degrees to the rolling direction of the wheel (figure 6.3).

Table 6.1 outlines the advantages and disadvantages of each wheel type.

Method	*Advantages*	*Disadvantages*
Omni Wheel	• Zero turning circle • Only requires three driven wheels • Can instantaneously drive in any direction	• Complex kinematics • Poor drive efficiency ~50% [24] • Expensive
Tradition wheel	• High drive efficiency • Cost-effective • Requires on 2 driven wheels • Simple to implement kinematics	• Large turning circle • Cannot instantaneously go in any direction
Mecanum Wheel	• Zero turning circle • Up to 50% higher drive efficiency than omni wheels [24] • Can instantaneously drive in any direction	• Requires four driven wheels • Complex kinematics • Expensive

Table 6.1: Alternative wheel types

Figure 6.1: Omni Wheels [31]

Figure 6.2: Traditional Wheels [30]

Figure 6.3: Mecanum Wheels [32]

After considering the advantages and disadvantages of the various wheel types it was decided that the mecanum wheel would be most suitable for the project. This is because of its high drive efficiency and ability to instantaneously drive in any direction. This will allow the rover to easily manoeuvre around the course and make small movements to change sensor position. This should help to offset the higher cost of the mecanum wheel as it negates the need for an expensive articulating to adjust sensor position.

6.2 Design Specification

Requirements of the chassis design and motor control.

Feature	Quantitative specification	Qualitative specification
Fast	Rover must be able to get from one side of the testing area to the other in under 15 seconds.	The rover can quickly travel between aliens and point at the alien's mouth.
Small Turning Circle	Must have a turning circle of less than 30 cm.	The rover can effectively navigate into small spaces and adjust to reposition sensors.
Light	The chassis and motors must weigh less than 500g.	To ensure that the rover does not trip the weight sensor.
Good value for money	The chassis must cost less than £25.	To leave enough money left for the rest of the rover and to allow the rover to present a good value.

Table 6.2: Chassis Design - Design Specification

6.3 Experimental Method

6.3.1 First Prototype

To start working on the kinematics of driving the mecanum a basic first prototype of the rover was 3D printed. Figures 6.4 and 6.5 show the initial prototype of the rover.

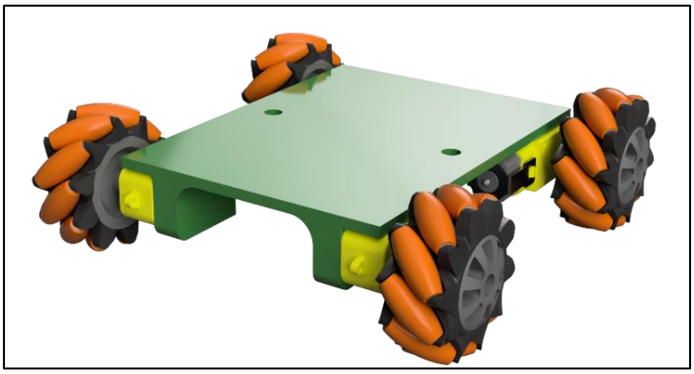

Figure 6.4: 3D render of first prototype

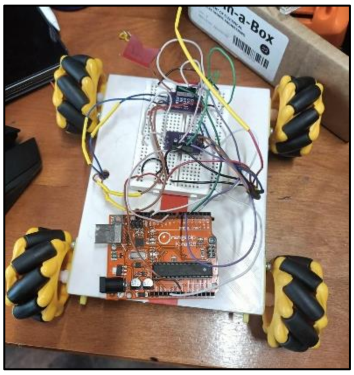

Figure 6.5: First prototype

6.3.2 Analysis of First Prototype

Relating to the design specification (table 6.2), the first prototype met the initial qualitative specifications. When tested, the rover drove well and was able to manoeuvre effectively around the course. Furthermore, the rover met the speed requirements and was below the weight limit when tested on the scales in the arena.

However, from the prototype, certain areas for improvement were highlighted. Table 6.2 indicates the problems faced and suggested solutions.

Areas for improvement	Solution
While driving the rover around the course, sometimes some connections on the breadboard would fall out.	Solder the circuit onto a Veroboard.
The mechanism to secure batteries was weak causing the battery pack to fall off.	Use brass threaded inserts to secure the battery packs.
The rover will drift when driving in a straight line.	Improve weight distribution to ensure none of the wheels slip.

Table 6.3: Improvements and solutions to first prototype

6.3.3 Inverse Kinematics of Driving Mecanum Wheels

Figure 6.6 shows the calculations for the inverse kinematics of the mecanum wheels. This will allow the rover to drive at a velocity, v with an angular velocity, ω.

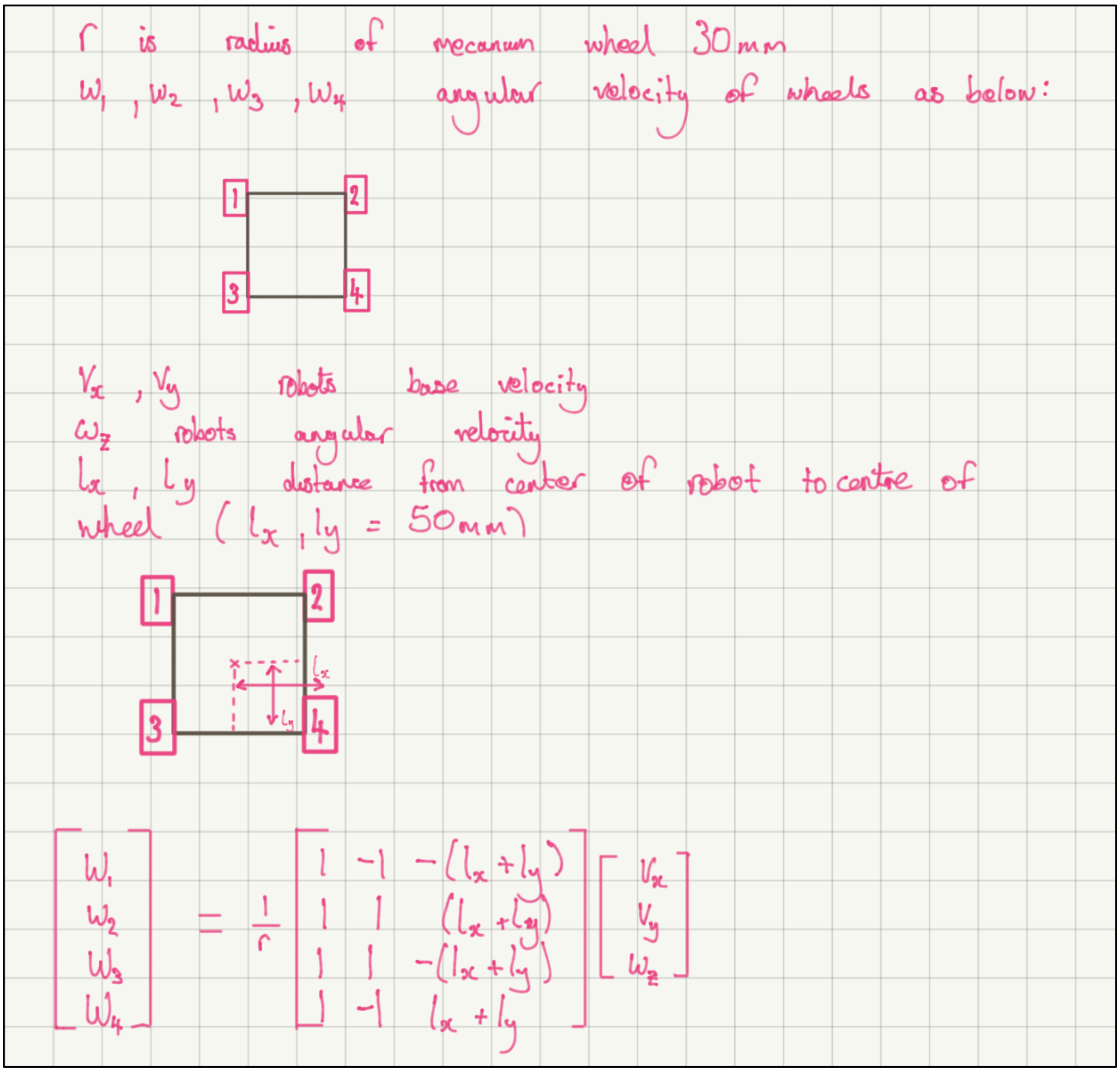

Figure 6.5: Calculations for the inverse kinematics of the mecanum wheels

The equation comes from the International Journal of Computer Applications Volume 113 2015 ‘Kinematic Model of a Four Mecanum Wheeled Mobile Robot’ Hamid Taheri, Bing Qiao & Nurallah Ghaeminezhad

6.3.4 Second Prototype

The second prototype was designed to resolve the issues with the original prototype and allow the sensors to get close enough to the alien to function. The second prototype also swaps the heavy AA batteries for lighter AAA batteries to save weight. Two versions were designed one with an articulating arm controlled by a servo (Figure 6.6) and one with a fixed coil height (Figure 6.7).

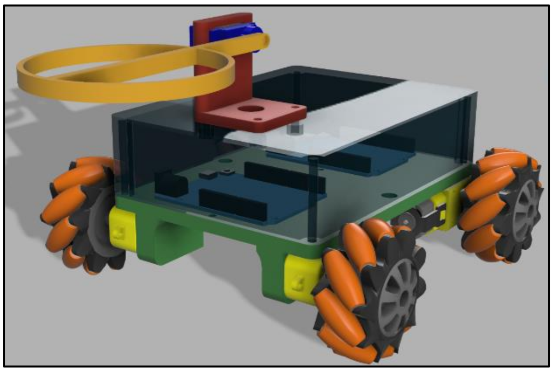

Figure 6.6: Second Prototype with articulating arm

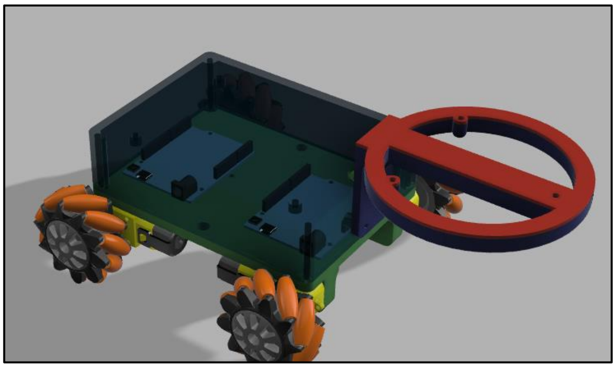

Figure 6.7: Second Prototype with fixed arm

After consultation with the relevant sub-teams, it was agreed that an articulating arm would only add weight cost to the design as the required sensor position manipulation could be achieved with mecanum wheels alone.

6.3.5 Final Design

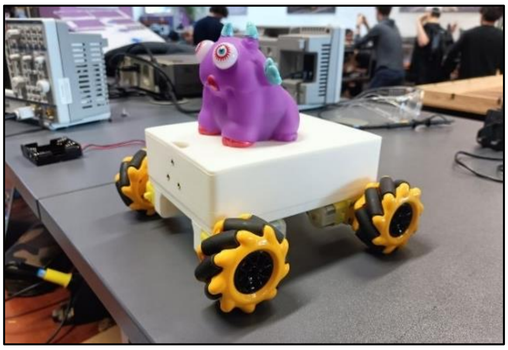

Figure 6.8: Final Design before arm is attached

Figure 6.9: Final Design with arm is attached

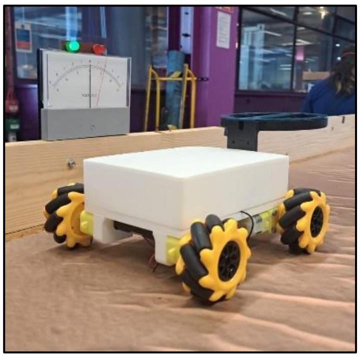

Figure 6.9: Weight-limit

The final circuitry for each sensor section will be combined on a single Veroboard. This is cheaper than customised PCB yet more reliable than using breadboards to eliminate loose connections.

6.4 Testing

Weight:

From Figure 6.9, the chassis of the rover with most of the internal circuitry did not exceed the weight limit due to a green light. Hence, it meets the specification for being lightweight.

Speed:

The rover travelled at a speed of approximately 1m/s when fully loaded which is sufficient to traverse the arena quickly on the demo day. It was able to navigate from one side of the arena to the other is well under the 15 second target.

Manoeuvrability:

When testing the complete movements of the chassis, the design of the mecanum wheels allowed the rover to turn in any direction without having to move from that point. This makes the rover very manoeuvrable satisfying the specification.

6.5 Evaluation

Overall, the chassis and motor design where a success as they meet all requirements set out in the specification – weight, speed, and manoeuvrability. The first prototype had several issues such as drift and a poor battery mounting mechanism. The use of rapid prototyping techniques such as 3D printing allowed the quick iteration of the design to allow for rapid improvement. Ultimately making our final design a success.

7. Integration of Submodules

After development on an OrangePip microcontroller, the magnetic field and age detection algorithms were tested on the Adafruit Metro M0 Express microcontroller. The range and accuracy of both saw itself largely decreased, due to the poor connection of the Wi-Fi shield's pins to the Adafruit board, generating noise, and the much larger sensitivity of the Adafruit's superior processor, picking up much more noise. Possible solutions to this problem are shown in table 7.1.

Method	*Advantages*	*Disadvantages*
Improve filtering and noise stabilization	• Makes circuit compact and lighter • Adaptable	• Expensive and time consuming • Requires sensing under 10ms • Requires new innovation
Using both OrangePip and Adafruit microcontrollers	• Can run sensing and motor driving in parallel • No need to modify existing systems • Simple, cheap and time efficient	• Takes up more space and weight • More complicated code Can cause syncing issues

Table 7.1: Alternative integration methods

As the second option was chosen due to its simplicity, and ability to solve multiple problems at once, which serial connection method to use must be found. A number exist, and they are compared in table 7.2.

Method	*Advantages*	*Disadvantages*
I2C Protocol *[25] [25]*	• Simple and easy to implement • Built in acknowledgement mechanism • Slave acknowledges it has received the command • Low pin count	• Relatively slow • Limited distance between controllers
UART *[26]*	• High data transfer rate • Long distance communication	• No built in acknowledgement mechanism • Additional wiring • No addressing • RX pin used by name detection
SPI *[27]*	• Highest speed • Allows full duplex communication • I/O concurrently	• Separate select line for each slave • Slave cannot initiate communication • Clock synchronization is sometimes difficult

Table 7.2: Alternative methods

From these, I2C protocol was chosen due to its simplicity. Its disadvantages are not relevant for this project since the master and slave will be right next to each other in the rover. The microcontrollers must therefore be connected as seen in Figure 7.1 (the Arduino Uno is the slave, the Arduino Leonardo, the master) [28].

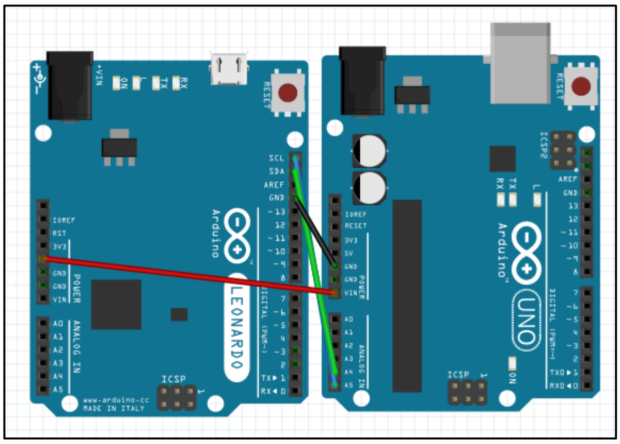

Figure 7.1: Combining Adafruit and Orangepip boards

<u>Development of Code</u>

After the individual programs for the age, magnetic and name detection were developed, they were combined together into a singular.ino file to be run on the OrangePip. This also needed to be set up as a slave device, for the Adafruit microcontroller to output. The structure of the code can therefore be seen in Figure 7.2.

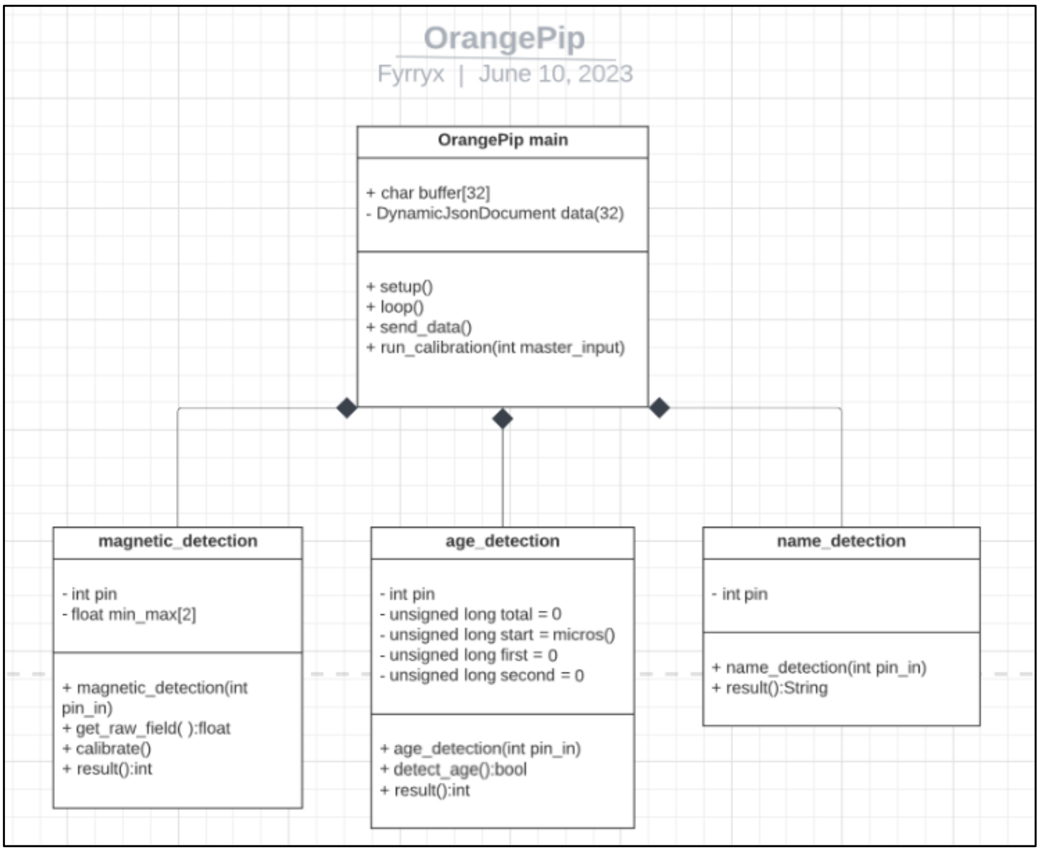

Figure 7.2 UML Interfacing Diagram

A class was created for each sensor, so that an object can be called with the appropriate data. The send_data() function is used to write to the master Adafruit device, and the run calibration() function is used to run the magnetic calibration function whenever the master device requests it.

8. Conclusion

This report has shown the design process of a rover which can remotely navigate a smooth arena with immovable obstacles and identify various characteristics of aliens located around the space, before communicating the obtained data to a suitable interface. Within each submodule of this overall task, various ways of meeting the necessary individual specifications were researched, tested, and evaluated. Subsequently, integration of the separate sections occurred. This included collating and appropriately modifying code for an integrated program, soldering sub-circuits to a Veroboard for a compact and unified overall circuit and combining hardware and software resulting in a completed rover.

The rover met the full design specification in appendix 3. The rover detected the name, age and magnetic field direction emitted by the alien whilst being remotely controlled by an XBOX controller plugged into the laptop. The data received was then displayed onto a graphical user interface. This allowed the rover to quickly continue its navigation of the arena as soon as data was obtained, without being forced to wait for information to be manually recorded. The chassis design allowed all the sensor circuitry to be enclosed without being exposed as per the specifications and was lightweight to ensure that the rover did not exceed the weight limit.

Had more time been allocated to this project, an improvement could have been to use servos to allow the 'sensing arm' of the chassis to move vertically and horizontally. This would have allowed for easier positional adjustment of the 'name', 'age' and 'magnetic field' sensors to receive a stronger signal. However, adding servos could have resulted in a weight above the limit. One idea that was considered for reducing weight was to use lithium batteries, however, these were identified as a potential fire-hazard.

Group Member Contributions

During the first group meeting, the project task was broken down and the workload was shared equally between the team members as shown in Table 7.1.

Name Detection	Katie Gloag, Sam Barber
Age Detection	Zian Lin, Kiertan Solanki
Magnetic Field Detection	Lolézio Viora Marquet
Remote Controlling and Data Output (I/O)	William Huynh
Chassis Design	Sam Barber, (Zian Lin)

Table 7.1: Submodule Responsibilities

A 'Fyrryx' GitHub, WhatsApp and Microsoft Teams were created to facilitate file sharing and communication. An initial Gantt Chart (Appendix 4) was created after initial research was conducted for each submodule. This was invaluable in guiding time management in the early stages of the project. Halfway through the project, an updated Gantt Chart (Appendix 5) was made which included more detail and an updated timescale. This allowed the team to reflect on progress and identify areas which required more attention as the project neared its end.

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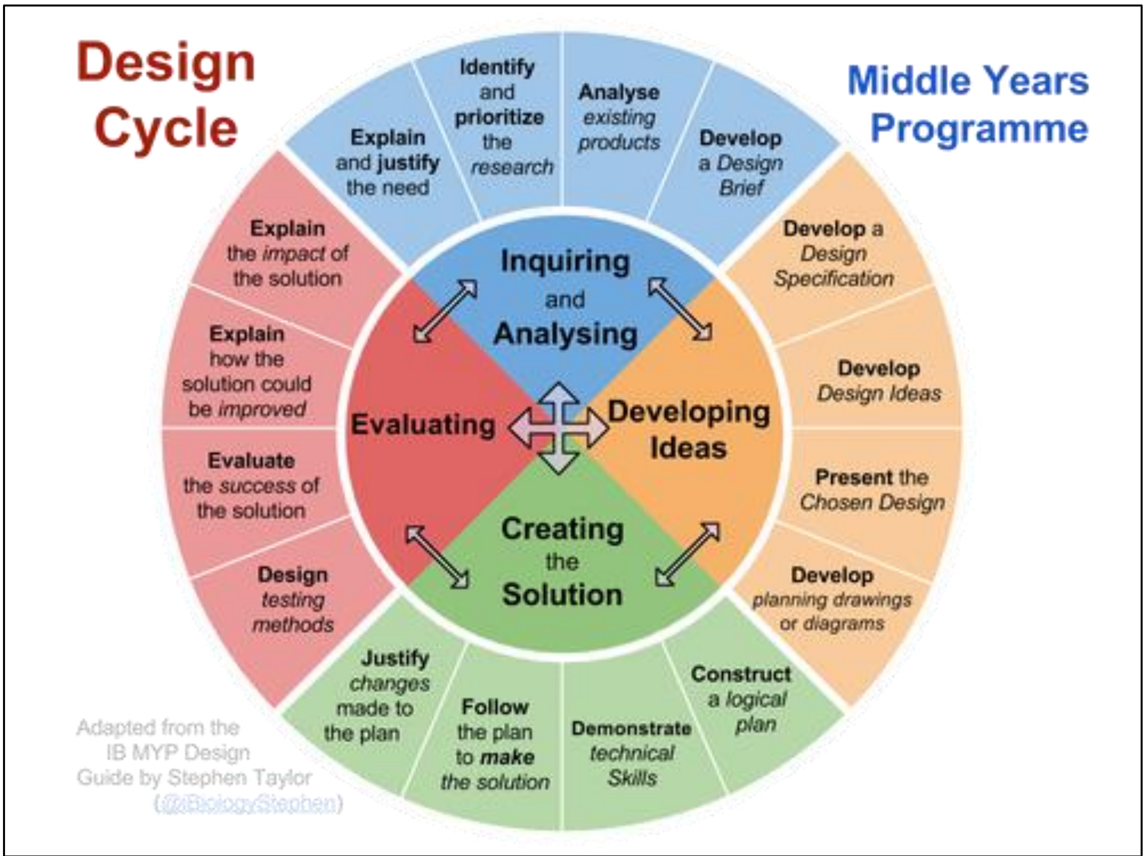
9. Appendix

Appendix 1: Design Cycle



Appendix 2: Finances:

Item	Quantity	Unit price (£)	Supplier	Date Purchased	Total spend (£)
QSD123 Phototransistor	1	0.62	Farnell	22/05/23	0.62
Hall effect sensor SS49E	5	0.68	eBay	22/05/23	4.02
Mecanum wheel 60mm	4	2.93	AliExpress	-	15.74
Piezo buzzer	2	0.82	Farnell	01/06/23	16.56
Veroboard	2	1.44	EE Stores	06/06/23	19.44
AD8056ANZ OpAmp	1	5.66	Farnell	06/06/23	25.10
74HCT14 SCHMITT TRIGGER	1	0.41	EE stores	06/06/23	25.51
Phototransistor 850 nm	2	0.41	Farnell	06/06/23	26.33
AAA battery	8	0.41	EE stores	06/06/23	29.61
AAA battery holder	2	1.11	RS	06/06/23	31.83
SCREWS BRASS M2 X 10MM	4	0.05	EE stores	12/06/23	32.03

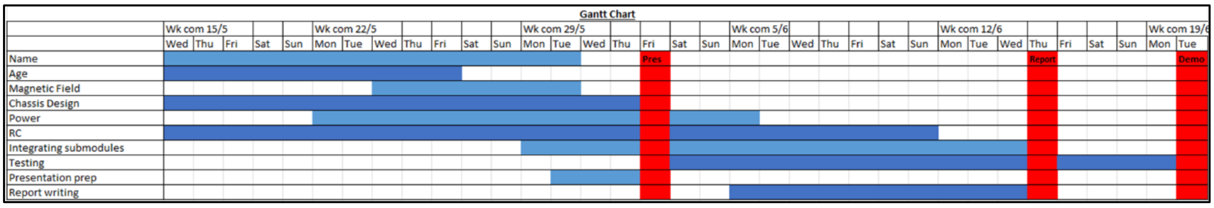
Total expenditure (£)	Remaining budget (£)
32.03	27.97

Appendix 3: Full 32-Point Design Specification

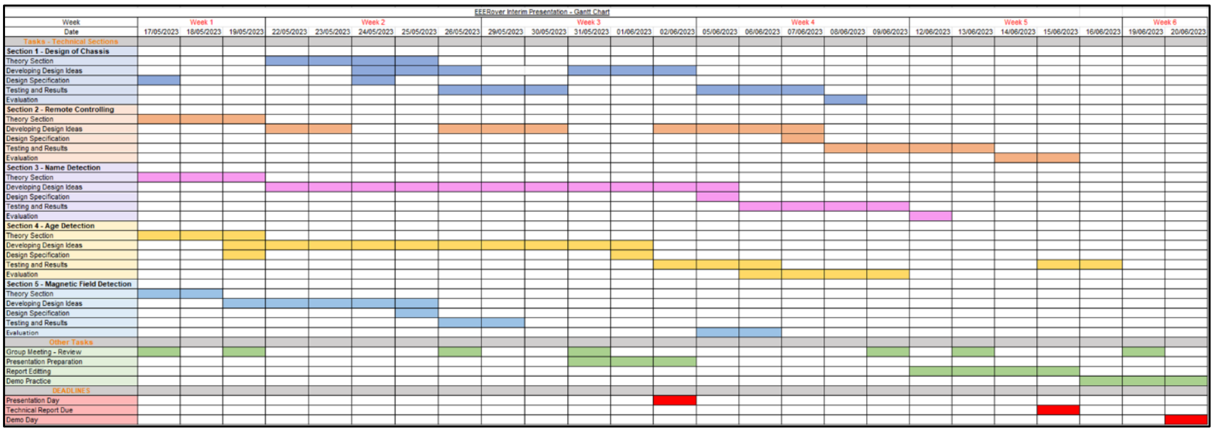
Specification	Explanation
Performance	• Manoeuvre the arena accurately and quickly • Identify alien name, age and magnetic field using radio, IR and magnetic field sensors • Continuously take movement input from the controller and return sensor output to the computer using Wi-Fi • Battery-powered using 14V • Max voltage to Metro board input is 3.3V
Environment	• Smooth ground - requires grip on wheels to prevent slipping • Large immovable obstacles - good handling and small turning circle for fast traversal across the arena • Room temperature, pressure, humidity - same conditions as testing • Newly built - contributes to smooth terrain, unlikely to have structural defects • Weight sensor - requires weight 'less than twice that of a reasonable rover' • Aliens situated >500mm apart - acceleration may be more valuable than speed • Misleading frequencies emitted by obstacles - small range of sensors
Life in Service	• Expected active run time is 3-4 minutes
Maintenance	• No maintenance requirements for the user • Easy access to maintenance as the environment is within the lab, and the demo will occur with all team members present • Reliability should be highly valued to reduce the frequency of maintenance • Spares of essential parts should be ordered in case of long delivery times when a spare is needed • All tools needed for maintenance are available in the lab
Target Product Cost	• Overall budget of £60 • Contingency budget of £15 • Cost efficiency valued - is performance increased enough to justify purchase
Competition	• Other rovers in arena • Will be competing for speed of completion • Protection against sabotage (e.g., other rovers emitting misleading IR signals) - small sensor range • If more than one rover is in the weight sensor area, the alien will not emit its signals - keep distance from other teams, especially in this region
Shipping	• Not applicable - product will be built, tested and demoed in the same lab
Packing	• Not applicable - product will be built, tested and demoed in the same lab
Quantity	• One rover is required in the brief • Prototypes can be built to test designs before committing
Manufacturing Facilities	• Lab facilities • Lab tech assistance is available • 3D printers and laser cutting available
Size	• Limited by weight restrictions and manoeuvrability • Aim for < 25x25cm
Weight	• < 'weight of two reasonable rovers' • Can be checked on the weight sensor • Weight needs to be well-balanced for the best handling, or else the rover could drift

Aesthetics, Appearance and Finish	• Rover aesthetics are a secondary concern - no customer requirements, but neat circuitry has benefits including 1) reducing the risk of loose connections, 2) ease of maintenance if problems arise • 'Logical' and 'easy to use' remote control interface - GUI for incoming data
Materials	• Chassis options: 3D printed PLA - cheaper than laser cut acrylic and lightweight • Circuitry: Veroboard - cheaper than customised PCB but more reliable than using breadboards as can eliminate loose connections • Wheels - Mecanum multi-directional wheels - increase manoeuvrability: turning circle of 0, can move sideways - good grip
Product Life Span	• One-time production • Life span > 20th June (demo date)
Standard and Specifications	• Battery safety
Ergonomics	• Ensuring a good user experience • Low latency • Good handling • Appealing GUI
Customer	• Age - adult, average motor skills • Height and strength should not affect the user experience
Quality and Reliability	• Motor max voltage = 57V, using 14V so well in-range • Sensors extensively tested individually - range, noise tolerance
Shelf Life and Storage	• Not applicable
Processes	• Not applicable - no special manufacturing processes
Time Scale	• 5 weeks for the project • 20th June demo date • Interim presentation 2nd June • Report due 15th June
Safety	• Use batteries with care • Labels and warnings are not applicable, as the rover will not be distributed • Do not inhale solder fumes
Company Constraints	• Not applicable - same as customer requirements (e.g., budget etc.)
Market Constraints	• Not applicable
Parents, Literature and Product Data	• Not applicable
Political and Social Implications	• Not applicable
Legal	• Not applicable - product will not be released to the general population
Installation	• None required - demo will be in the lab
Documentation	• Report sent to client detailing design process and general project progress • Circuit diagrams, simulations and videos are to be attached to the report • References are to be detailed in the appendix of the report
Disposal	• Rover can be dismantled to reuse parts after use • Batteries should be disposed of at a suitable facility

Appendix 4:



Appendix 5



Appendix 6: GitHub Link

https://github.com/saturn691/Fyrryx